Mathematical Statistics

Yifan Li

August 2026

Contents

Introduction **4**

1 Probability and Distributions **5**
1.1 Expectation, Variance, and Moments . . . 6
1.2 Quantiles and Medians . . . 8
1.3 Basic Probability Inequalities . . . 8
1.4 Moment Generating Functions . . . 9
1.5 Common Distributions . . . 9
1.5.1 Bernoulli and Binomial Distributions . . . 9
1.5.2 Poisson Distribution . . . 10
1.5.3 Geometric and Negative Binomial Distributions . . . 10
1.5.4 Uniform Distribution . . . 11
1.5.5 Beta Distribution . . . 11
1.5.6 Exponential Distribution . . . 11
1.5.7 Gamma Distribution . . . 11
1.5.8 Normal Distribution . . . 12
1.5.9 Chi-Square Distribution . . . 12
1.6 Joint Distributions . . . 13
1.7 Conditional Distribution and Conditional Expectation . . . 14
1.8 Bivariate Normal Distribution . . . 14
1.9 Transformations of Random Variables . . . 15
1.10 Inverse Bias Formula . . . 15
1.11 Zero-Truncated Poisson Distribution . . . 16

2 Sample Distributions **17**
2.1 Functions of Random Samples . . . 18
2.2 Sample Mean and Sample Variance . . . 19
2.3 Sampling from a Normal Population . . . 20
2.4 Student's t Distribution . . . 22
2.5 F Distribution . . . 23
2.6 Order Statistics . . . 24
2.6.1 Distribution of the Maximum . . . 24
2.6.2 Distribution of the Minimum . . . 24
2.6.3 General Order Statistic . . . 25
2.7 Law of Large Numbers . . . 25
2.8 Central Limit Theorem . . . 26
2.9 Normal Approximation to the Binomial Distribution . . . 27
2.10 Poisson Approximation to the Binomial Distribution . . . 28
2.11 Summary of Important Sampling Distributions . . . 28

3 Point Estimation **30**
3.1 Basic Concepts of Point Estimation . . . 31
3.2 Method of Moments . . . 31
3.3 Maximum Likelihood Estimation . . . 33
3.3.1 Example: Bernoulli Model . . . 34
3.3.2 Example: Poisson Model . . . 35

3.3.3 Example: Normal Model . . . 35
3.4 Invariance Property of the MLE . . . 36
3.5 Bayesian Point Estimation . . . 36
3.6 Bias and Unbiased Estimation . . . 37
3.7 Mean Squared Error . . . 38
3.8 Consistency . . . 39
3.9 Score Function and Fisher Information . . . 39
3.10 Cramér–Rao Inequality . . . 41
3.11 Uniformly Minimum Variance Unbiased Estimation . . . 42
3.12 Sufficient Statistics . . . 43
3.13 Factorization Theorem . . . 44
3.14 Rao–Blackwell Improvement . . . 45
3.15 Complete Statistics . . . 46
3.16 Lehmann–Scheffé Theorem . . . 47
3.17 Example: Poisson UMVUE . . . 48
3.18 Relationships Among Estimation Concepts . . . 48
3.19 Summary of Point Estimation . . . 49

4 Confidence Interval Estimation 50
4.1 Introduction to Confidence Intervals . . . 51
4.2 Pivotal Quantities . . . 51
4.3 Quantiles . . . 52
4.4 Confidence Interval for a Normal Mean . . . 53
4.4.1 Variance Known . . . 53
4.4.2 Variance Unknown . . . 53
4.5 Confidence Interval for the Difference of Two Normal Means . . . 54
4.5.1 Variances Known . . . 54
4.5.2 Variances Unknown but Equal . . . 54
4.6 Confidence Interval for a Normal Variance . . . 55
4.6.1 Mean Known . . . 55
4.6.2 Mean Unknown . . . 56
4.7 Confidence Interval for the Ratio of Two Normal Variances . . . 56
4.8 Large-Sample Confidence Intervals . . . 57
4.8.1 General Asymptotic Normal Interval . . . 57
4.8.2 Large-Sample Interval Based on an MLE . . . 58
4.9 Large-Sample Confidence Interval for a Population Proportion . . . 58
4.10 Sample Size Determination . . . 59
4.10.1 Normal Mean with Known Variance . . . 59
4.10.2 Population Proportion . . . 59
4.11 Summary of Confidence Intervals . . . 60

5 Hypothesis Testing 61
5.1 Introduction to Hypothesis Testing . . . 62
5.2 Critical Region and Decision Rule . . . 62
5.3 Type I and Type II Errors . . . 63
5.4 Power Function, Size, and Significance Level . . . 64
5.5 The p-Value . . . 64
5.6 Neyman–Pearson Lemma . . . 65
5.7 Example of the Neyman–Pearson Lemma . . . 65

5.8 Likelihood Ratio Test . . . 66
5.9 Procedure for Constructing an LRT . . . 67
5.10 Asymptotic Distribution of the Likelihood Ratio . . . 67
5.11 Tests for a Normal Mean . . . 68
5.12 One-Sample Z Test: Variance Known . . . 68
5.12.1 Upper-Tailed Test . . . 68
5.12.2 Lower-Tailed Test . . . 68
5.12.3 Two-Sided Test . . . 68
5.13 One-Sample t Test: Variance Unknown . . . 69
5.14 Two-Sample Test for Normal Means . . . 69
5.14.1 Variances Known . . . 70
5.14.2 Variances Unknown but Equal . . . 70
5.15 Connection Between Confidence Intervals and Tests . . . 70
5.16 Chi-Square Goodness-of-Fit Test . . . 71
5.17 Pearson's Chi-Square Statistic . . . 72
5.18 Completely Specified Distribution . . . 72
5.19 Example: Discrete Uniform Distribution . . . 73
5.20 Goodness-of-Fit with Estimated Parameters . . . 73
5.21 Example: Testing a Poisson Distribution . . . 74
5.22 Combining Categories . . . 74
5.23 Pearson Statistic as a Measure of Discrepancy . . . 75
5.24 Summary of Hypothesis Testing . . . 75

Acknowledgements **77**

Introduction

Mathematical statistics studies how information about an unknown population can be extracted from observed random data. While probability theory starts with a known probability model and studies the random outcomes generated by that model, mathematical statistics proceeds in the opposite direction: we observe the outcomes and use them to infer the underlying model.

Suppose that

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x;\theta),$$

where θ is an unknown parameter. The random variables $X_1, \ldots, X_n$ form a *random sample* from the population, while their observed values

$$x_1, \ldots, x_n$$

constitute the observed data.

A *statistic* is a function

$$T = T(X_1, \ldots, X_n)$$

of the sample that does not involve unknown parameters. Statistics provide the basic tools for extracting information from the sample. Typical examples include the sample mean

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

and the sample variance

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2.$$

The basic structure of statistical inference can therefore be summarized as

$$\boxed{\text{Population} \longrightarrow \text{Random Sample} \longrightarrow \text{Statistic} \longrightarrow \text{Inference.}}$$

Three fundamental types of statistical inference will be studied.

- **Point estimation:** construct a statistic $\widehat{\theta}$ to estimate an unknown parameter θ, and study its bias, variance, mean squared error, consistency, and efficiency.
- **Confidence interval estimation:** construct a random interval

  $$[L(X_1, \ldots, X_n),\, U(X_1, \ldots, X_n)]$$

  that contains the unknown parameter with a prescribed confidence level.
- **Hypothesis testing:** use the observed sample to decide between competing hypotheses such as

  $$H_0 : \theta = \theta_0 \qquad \text{and} \qquad H_1 : \theta \neq \theta_0.$$

A central theme throughout the course is the *sampling distribution*. Since a statistic is itself a random variable, understanding its distribution allows us to quantify the uncertainty caused by random sampling. Important examples include the normal, chi-square, t, and F distributions, which form the basis of many classical estimation and testing procedures.

The course will proceed through the following chain:

$$\boxed{\text{Probability and Distributions} \longrightarrow \text{Sampling Distributions} \longrightarrow \text{Point Estimation} \longrightarrow \text{Confidence Interv}}$$

The main objective is not merely to collect statistical formulas, but to understand how each formula arises from the underlying probability model and how it is used to make inference from random data.

Chapter 1

Probability and Distributions

1 Probability and Distributions

1.1 Expectation, Variance, and Moments

Probability theory provides the mathematical foundation for statistical inference. In particular, expectation and variance describe the location and dispersion of a random variable and will later be used to evaluate estimators.

Definition 1.1 (Expectation)**.** Let X be a random variable.

If X is discrete with probability mass function $p_X(x)$, then

$$\mathbb{E}[X] = \sum_x x\, p_X(x),$$

provided that the sum exists.

If X is continuous with probability density function $f_X(x)$, then

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\ \mathrm{d}x,$$

provided that the integral exists.

More generally, for a function g,

$$\mathbb{E}[g(X)] = \sum_x g(x) p_X(x)$$

in the discrete case, and

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\ \mathrm{d}x$$

in the continuous case.

Remark 1.2. The formula for $\mathbb{E}[g(X)]$ does not require finding the distribution of $g(X)$ first. This is often called the *law of the unconscious statistician*.

Proposition 1.3 (Linearity of Expectation)**.** *For constants $a_1, \ldots, a_n$,*

$$\mathbb{E}\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i \mathbb{E}[X_i].$$

In particular,

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b.$$

Remark 1.4. Independence is *not* required for linearity of expectation.

Definition 1.5 (Variance)**.** The variance of X is

$$\mathrm{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right].$$

Equivalently,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \Big(\mathbb{E}[X]\Big)^2.$$

For constants a, b,

$$\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X).$$

Definition 1.6 (Covariance)**.** For random variables X and Y,

$$\operatorname{Cov}(X,Y) = \mathbb{E}\left[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\right].$$

Equivalently,

$$\operatorname{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

The variance of a linear combination satisfies

$$\operatorname{Var}\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i^2 \operatorname{Var}(X_i) + 2\sum_{i<j} a_i a_j \operatorname{Cov}(X_i, X_j).$$

If $X_1, \ldots, X_n$ are independent, then

$$\operatorname{Cov}(X_i, X_j) = 0, \qquad i \neq j,$$

and hence

$$\operatorname{Var}\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i^2 \operatorname{Var}(X_i).$$

Definition 1.7 (Moment)**.** The k-th raw moment of X is

$$\mu_k' = \mathbb{E}[X^k].$$

The k-th central moment is

$$\mu_k = \mathbb{E}\left[(X - \mathbb{E}[X])^k\right].$$

In particular,

$$\mu_1' = \mathbb{E}[X], \qquad \mu_2 = \operatorname{Var}(X).$$

Two important standardized moments are the coefficient of skewness

$$\gamma_1 = \frac{\mu_3}{\sigma^3}$$

and the coefficient of kurtosis

$$\beta_2 = \frac{\mu_4}{\sigma^4},$$

where

$$\sigma^2 = \operatorname{Var}(X).$$

The excess kurtosis is

$$\gamma_2 = \beta_2 - 3.$$

Remark 1.8. Skewness measures asymmetry, while kurtosis is related to the heaviness of the tails and the concentration of probability around the center.

The normal distribution has

$$\gamma_1 = 0, \qquad \beta_2 = 3, \qquad \gamma_2 = 0.$$

1.2 Quantiles and Medians

Definition 1.9 (Quantile)**.** Let F be the cumulative distribution function of X. For $0 < p < 1$, a p-quantile is a value q_p satisfying, in the continuous strictly increasing case,

$$F(q_p) = p.$$

Equivalently,

$$q_p = F^{-1}(p).$$

The median is the 0.5-quantile:

$$\mathrm{Med}(X) = q_{0.5}.$$

For a symmetric distribution about μ,

$$\mathrm{Med}(X) = \mu$$

whenever the median is unique.

Remark 1.10. The mean and median describe different notions of location. The mean is sensitive to extreme observations, whereas the median is more resistant to outliers.

For a symmetric distribution they often coincide, but this is not true for a general distribution.

1.3 Basic Probability Inequalities

Several probability inequalities are repeatedly used to obtain bounds when the exact distribution of a random variable is unavailable.

Theorem 1.11 (Markov's Inequality)**.** *If $X \geq 0$ and $a > 0$, then*

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Theorem 1.12 (Chebyshev's Inequality)**.** *If $\mathbb{E}[X] = \mu$ and $\mathrm{Var}(X) = \sigma^2 < \infty$, then for every $a > 0$,*

$$\mathbb{P}(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}.$$

Equivalently, for $k > 0$,

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

Remark 1.13. Chebyshev's inequality requires only a finite variance and does not assume that X follows a normal distribution. Its generality makes it useful for proving convergence results, although the resulting bound may be loose.

Theorem 1.14 (Convexity Inequality)**.** *If g is a convex function and the expectations exist, then*

$$g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)].$$

Theorem 1.15 (Cauchy–Schwarz Inequality)**.** *If X and Y have finite second moments, then*

$$|\mathbb{E}[XY]|^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2].$$

Equality holds if and only if

$$Y = cX$$

almost surely for some constant c, provided neither variable is trivial.

1.4 Moment Generating Functions

Definition 1.16 (Moment Generating Function)**.** The moment generating function (MGF) of a random variable X is

$$M_X(t) = \mathbb{E}[e^{tX}],$$

for those values of t for which the expectation exists.

If $M_X(t)$ exists in a neighborhood of $t = 0$, then

$$M_X^{(k)}(0) = \mathbb{E}[X^k].$$

In particular,

$$M_X'(0) = \mathbb{E}[X], \qquad M_X''(0) = \mathbb{E}[X^2],$$

so that

$$\mathrm{Var}(X) = M_X''(0) - \Big(M_X'(0)\Big)^2.$$

Proposition 1.17 (MGF of an Affine Transformation)**.** *If*

$$Y = aX + b,$$

then

$$M_Y(t) = e^{bt} M_X(at).$$

Proposition 1.18 (MGF of a Sum)**.** *If $X_1, \ldots, X_n$ are independent, then*

$$M_{X_1+\cdots+X_n}(t) = \prod_{i=1}^{n} M_{X_i}(t).$$

Remark 1.19. This property makes MGFs particularly useful for determining the distribution of sums of independent random variables. If the resulting MGF can be recognized as the MGF of a known distribution, the distribution of the sum follows immediately.

1.5 Common Distributions

The following distributions occur repeatedly in mathematical statistics.

1.5.1 Bernoulli and Binomial Distributions

If

$$X \sim \mathrm{Bernoulli}(p),$$

then

$$\mathbb{P}(X = x) = p^x (1-p)^{1-x}, \qquad x = 0, 1.$$

Its mean and variance are

$$\mathbb{E}[X] = p, \qquad \mathrm{Var}(X) = p(1-p),$$

and its MGF is

$$M_X(t) = 1 - p + pe^t.$$

If

$$X \sim \mathrm{Binomial}(n, p),$$

then

$$\mathbb{P}(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, \ldots, n.$$

Moreover,

$$\mathbb{E}[X] = np, \qquad \text{Var}(X) = np(1-p),$$

and

$$M_X(t) = (1 - p + pe^t)^n.$$

A binomial random variable may be represented as

$$X = X_1 + \cdots + X_n,$$

where

$$X_i \overset{\text{iid}}{\sim} \text{Bernoulli}(p).$$

1.5.2 Poisson Distribution

If

$$X \sim \text{Poisson}(\lambda), \qquad \lambda > 0,$$

then

$$\mathbb{P}(X = x) = e^{-\lambda}\frac{\lambda^x}{x!}, \qquad x = 0, 1, 2, \ldots.$$

Its mean and variance are

$$\mathbb{E}[X] = \text{Var}(X) = \lambda,$$

and

$$M_X(t) = \exp\{\lambda(e^t - 1)\}.$$

If

$$X_i \sim \text{Poisson}(\lambda_i), \qquad i = 1, \ldots, n,$$

are independent, then

$$\sum_{i=1}^{n} X_i \sim \text{Poisson}\left(\sum_{i=1}^{n} \lambda_i\right).$$

1.5.3 Geometric and Negative Binomial Distributions

For the convention

$$X = \text{number of trials until the first success},$$

if

$$X \sim \text{Geometric}(p),$$

then

$$\mathbb{P}(X = x) = (1-p)^{x-1}p, \qquad x = 1, 2, \ldots.$$

Its mean and variance are

$$\mathbb{E}[X] = \frac{1}{p}, \qquad \text{Var}(X) = \frac{1-p}{p^2}.$$

The negative binomial distribution generalizes this idea to the number of trials required to obtain a fixed number of successes.

1.5.4 Uniform Distribution

If

$$X \sim U(a, b),$$

then

$$f_X(x) = \frac{1}{b-a}, \qquad a < x < b.$$

Its mean and variance are

$$\mathbb{E}[X] = \frac{a+b}{2}, \qquad \mathrm{Var}(X) = \frac{(b-a)^2}{12}.$$

1.5.5 Beta Distribution

If

$$X \sim \mathrm{Beta}(\alpha, \beta), \qquad \alpha, \beta > 0,$$

then

$$f_X(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1}(1-x)^{\beta-1}, \qquad 0 < x < 1,$$

where

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}.$$

Its mean and variance are

$$\mathbb{E}[X] = \frac{\alpha}{\alpha+\beta},$$

and

$$\mathrm{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.$$

1.5.6 Exponential Distribution

If

$$X \sim \mathrm{Exponential}(\lambda), \qquad \lambda > 0,$$

then

$$f_X(x) = \lambda e^{-\lambda x}, \qquad x > 0.$$

Its mean and variance are

$$\mathbb{E}[X] = \frac{1}{\lambda}, \qquad \mathrm{Var}(X) = \frac{1}{\lambda^2}.$$

The exponential distribution satisfies the memoryless property

$$\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t).$$

1.5.7 Gamma Distribution

Using the shape–rate parameterization,

$$X \sim \mathrm{Gamma}(\alpha, \lambda),$$

has density

$$f_X(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \qquad x > 0.$$

Then

$$\mathbb{E}[X] = \frac{\alpha}{\lambda}, \qquad \mathrm{Var}(X) = \frac{\alpha}{\lambda^2},$$

and

$$M_X(t) = \left(\frac{\lambda}{\lambda - t}\right)^{\alpha}, \qquad t < \lambda.$$

If

$$X_i \sim \mathrm{Gamma}(\alpha_i, \lambda)$$

are independent and have the same rate λ, then

$$\sum_{i=1}^{n} X_i \sim \mathrm{Gamma}\left(\sum_{i=1}^{n} \alpha_i, \lambda\right).$$

1.5.8 Normal Distribution

If

$$X \sim N(\mu, \sigma^2),$$

then

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}, \qquad -\infty < x < \infty.$$

Its mean and variance are

$$\mathbb{E}[X] = \mu, \qquad \mathrm{Var}(X) = \sigma^2,$$

and its MGF is

$$M_X(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right).$$

If

$$X \sim N(\mu, \sigma^2),$$

then the standardized variable

$$Z = \frac{X - \mu}{\sigma}$$

satisfies

$$Z \sim N(0, 1).$$

If

$$X_i \sim N(\mu_i, \sigma_i^2)$$

are independent, then

$$\sum_{i=1}^{n} a_i X_i \sim N\left(\sum_{i=1}^{n} a_i \mu_i, \sum_{i=1}^{n} a_i^2 \sigma_i^2\right).$$

1.5.9 Chi-Square Distribution

If

$$Z_1, \ldots, Z_\nu \overset{\text{iid}}{\sim} N(0, 1),$$

then

$$X = \sum_{i=1}^{\nu} Z_i^2$$

has a chi-square distribution with ν degrees of freedom:

$$X \sim \chi^2_\nu.$$

Its mean and variance are

$$\mathbb{E}[X] = \nu, \qquad \operatorname{Var}(X) = 2\nu.$$

The chi-square distribution is a special case of the gamma distribution:

$$\chi^2_\nu \overset{\mathrm{d}}{=} \operatorname{Gamma}\left(\frac{\nu}{2}, \frac{1}{2}\right)$$

under the shape–rate parameterization.

Remark 1.20. The chi-square distribution will later play a central role in inference for normal variances and in chi-square goodness-of-fit tests.

1.6 Joint Distributions

Definition 1.21 (Joint Distribution)**.** For two discrete random variables X and Y, the joint probability mass function is

$$p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y).$$

For continuous random variables, the joint density $f_{X,Y}(x, y)$ satisfies

$$\mathbb{P}\Big((X, Y) \in A\Big) = \iint_A f_{X,Y}(x, y)\ \mathrm{d}x\ \mathrm{d}y.$$

The marginal densities are obtained by integrating out the other variable:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\ \mathrm{d}y,$$

and

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\ \mathrm{d}x.$$

Definition 1.22 (Independence)**.** Random variables X and Y are independent if

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

for all relevant x, y in the continuous case, with the analogous factorization for probability mass functions in the discrete case.

Definition 1.23 (Correlation)**.** If X and Y have finite positive variances, their correlation coefficient is

$$\rho_{X,Y} = \frac{\operatorname{Cov}(X, Y)}{\sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}}.$$

By the Cauchy–Schwarz inequality,

$$-1 \le \rho_{X,Y} \le 1.$$

If X and Y are independent, then

$$\operatorname{Cov}(X, Y) = 0.$$

The converse is not true in general.

Example 1.24 (Uncorrelated Does Not Imply Independent)**.** Let

$$X \sim U(-1,1), \qquad Y = X^2.$$

By symmetry,

$$\mathbb{E}[X] = 0, \qquad \mathbb{E}[XY] = \mathbb{E}[X^3] = 0.$$

Therefore

$$\mathrm{Cov}(X,Y) = 0.$$

However, $Y = X^2$ is completely determined by X, so X and Y are not independent.

1.7 Conditional Distribution and Conditional Expectation

For continuous random variables,

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)},$$

provided that $f_Y(y) > 0$.

The conditional expectation is

$$\mathbb{E}[X \mid Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x \mid y)\, \mathrm{d}x.$$

Theorem 1.25 (Law of Total Expectation)**.** *If the relevant expectations exist, then*

$$\mathbb{E}[X] = \mathbb{E}\left[\mathbb{E}[X \mid Y]\right].$$

Theorem 1.26 (Law of Total Variance)**.** *If X has finite variance, then*

$$\mathrm{Var}(X) = \mathbb{E}\left[\mathrm{Var}(X \mid Y)\right] + \mathrm{Var}\left(\mathbb{E}[X \mid Y]\right).$$

Remark 1.27. The total variance formula decomposes the total uncertainty in X into

$$\boxed{\text{average uncertainty within groups} + \text{uncertainty between groups.}}$$

This decomposition appears repeatedly in statistics.

1.8 Bivariate Normal Distribution

A pair (X,Y) has a bivariate normal distribution if its joint density can be written as

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_X)^2}{\sigma_X^2} - \frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \frac{(y-\mu_Y)^2}{\sigma_Y^2}\right]\right\}.$$

The marginal distributions are

$$X \sim N(\mu_X, \sigma_X^2), \qquad Y \sim N(\mu_Y, \sigma_Y^2),$$

with

$$\mathrm{Cov}(X,Y) = \rho\sigma_X\sigma_Y.$$

An important special property of jointly normal random variables is

$$\mathrm{Cov}(X,Y) = 0 \quad \Longrightarrow \quad X \text{ and } Y \text{ are independent.}$$

Remark 1.28. For arbitrary random variables, zero covariance does not imply independence. The implication above is a special property of jointly normal random variables.

1.9 Transformations of Random Variables

Suppose

$$Y = g(X)$$

and g is one-to-one and differentiable. If

$$x = g^{-1}(y),$$

then the density of Y is

$$f_Y(y) = f_X\Big(g^{-1}(y)\Big)\left|\frac{\mathrm{d}}{\mathrm{d}y}g^{-1}(y)\right|.$$

For multivariate transformations, suppose

$$Y_1 = g_1(X_1, X_2), \qquad Y_2 = g_2(X_1, X_2),$$

with inverse transformation

$$X_1 = h_1(Y_1, Y_2), \qquad X_2 = h_2(Y_1, Y_2).$$

Then

$$f_{Y_1,Y_2}(y_1, y_2) = f_{X_1,X_2}\Big(h_1(y_1, y_2), h_2(y_1, y_2)\Big)|J|,$$

where

$$J = \frac{\partial(x_1, x_2)}{\partial(y_1, y_2)} = \begin{vmatrix} \dfrac{\partial x_1}{\partial y_1} & \dfrac{\partial x_1}{\partial y_2} \\ \dfrac{\partial x_2}{\partial y_1} & \dfrac{\partial x_2}{\partial y_2} \end{vmatrix}.$$

Remark 1.29. Transformation techniques are particularly important in mathematical statistics because many sampling distributions are derived by transforming simpler random variables.

1.10 Inverse Bias Formula

Suppose g is sufficiently smooth and X is concentrated around its mean

$$\mu = \mathbb{E}[X].$$

A second-order Taylor expansion gives

$$g(X) \approx g(\mu) + g'(\mu)(X - \mu) + \frac{1}{2}g''(\mu)(X - \mu)^2.$$

Taking expectations,

$$\mathbb{E}[g(X)] \approx g(\mu) + \frac{1}{2}g''(\mu)\,\mathrm{Var}(X).$$

Thus, in general,

$$\mathbb{E}[g(X)] \neq g(\mathbb{E}[X]).$$

Remark 1.30. This approximation explains why applying a nonlinear transformation to an unbiased estimator generally introduces bias. It will be useful when studying transformed estimators and approximate inference.

1.11 Zero-Truncated Poisson Distribution

Suppose

$$X \sim \text{Poisson}(\lambda),$$

but observations with $X = 0$ cannot be observed. The observed random variable therefore follows the conditional distribution

$$X \mid X > 0.$$

For

$$x = 1, 2, \ldots,$$

we have

$$\mathbb{P}(X = x \mid X > 0) = \frac{\mathbb{P}(X = x)}{\mathbb{P}(X > 0)} = \frac{e^{-\lambda}\lambda^x/x!}{1 - e^{-\lambda}}.$$

Equivalently,

$$\mathbb{P}(X = x \mid X > 0) = \frac{\lambda^x}{x!(e^{\lambda} - 1)}, \qquad x = 1, 2, \ldots.$$

This distribution is called the *zero-truncated Poisson distribution*.

Remark 1.31. Truncation changes the probability model. Therefore one cannot simply use the ordinary Poisson likelihood when zero observations are systematically absent.

This example will later be revisited when maximum likelihood estimators must be obtained numerically.

Chapter 2

Sample Distributions

2 Sample Distributions

2.1 Functions of Random Samples

Let

$$X_1, \ldots, X_n$$

be a random sample from a population with distribution depending on a parameter θ.

A statistic

$$T = T(X_1, \ldots, X_n)$$

is itself a random variable. Its probability distribution is called the *sampling distribution* of T.

Definition 2.1 (Sampling Distribution)**.** The probability distribution of a statistic

$$T(X_1, \ldots, X_n)$$

under repeated random sampling is called the *sampling distribution* of T.

Remark 2.2. The central problem of this chapter is therefore:

$$\boxed{X_1, \ldots, X_n \sim f(x;\theta) \quad \Longrightarrow \quad \text{find the distribution of } T(X_1, \ldots, X_n).}$$

Sampling distributions form the probabilistic foundation for confidence intervals and hypothesis tests.

Suppose first that

$$Y = g(X),$$

where g is one-to-one and differentiable. If

$$x = g^{-1}(y),$$

then

$$f_Y(y) = f_X\Big(g^{-1}(y)\Big)\left|\frac{\mathrm{d}}{\mathrm{d}y}g^{-1}(y)\right|.$$

For a transformation of several random variables, the Jacobian method from Chapter 1 can be used.

For example, if

$$Y_1 = g_1(X_1, X_2), \qquad Y_2 = g_2(X_1, X_2),$$

and the inverse transformation is

$$X_1 = h_1(Y_1, Y_2), \qquad X_2 = h_2(Y_1, Y_2),$$

then

$$f_{Y_1,Y_2}(y_1, y_2) = f_{X_1,X_2}\Big(h_1(y_1, y_2), h_2(y_1, y_2)\Big)\left|\frac{\partial(x_1, x_2)}{\partial(y_1, y_2)}\right|.$$

2.2 Sample Mean and Sample Variance

Let

$$X_1, \ldots, X_n$$

be a random sample from a population with

$$\mathbb{E}[X_i] = \mu, \qquad \mathrm{Var}(X_i) = \sigma^2.$$

Definition 2.3 (Sample Mean)**.** The sample mean is

$$\overline{X} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

Definition 2.4 (Sample Variance)**.** The sample variance is

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \overline{X})^2.$$

The following identity is often useful:

$$\sum_{i=1}^{n} (X_i - \overline{X})^2 = \sum_{i=1}^{n} X_i^2 - n\overline{X}^2.$$

Therefore,

$$S^2 = \frac{1}{n-1} \left(\sum_{i=1}^{n} X_i^2 - n\overline{X}^2 \right).$$

Proposition 2.5 (Mean and Variance of the Sample Mean)**.** *If*

$$X_1, \ldots, X_n$$

are independent and identically distributed with

$$\mathbb{E}[X_i] = \mu, \qquad \mathrm{Var}(X_i) = \sigma^2,$$

then

$$\mathbb{E}[\overline{X}] = \mu$$

and

$$\mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n}.$$

Proof. By linearity of expectation,

$$\mathbb{E}[\overline{X}] = \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}[X_i] = \mu.$$

Since the observations are independent,

$$\mathrm{Var}(\overline{X}) = \mathrm{Var}\left(\frac{1}{n} \sum_{i=1}^{n} X_i \right) = \frac{1}{n^2} \sum_{i=1}^{n} \mathrm{Var}(X_i) = \frac{\sigma^2}{n}.$$

□

Theorem 2.6 (Unbiasedness of the Sample Variance)**.** *If*

$$X_1, \ldots, X_n$$

are iid with finite variance σ^2*, then*

$$\mathbb{E}[S^2] = \sigma^2.$$

Proof. Using

$$\sum_{i=1}^{n}(X_i - \overline{X})^2 = \sum_{i=1}^{n}(X_i - \mu)^2 - n(\overline{X} - \mu)^2,$$

we obtain

$$\mathbb{E}\left[\sum_{i=1}^{n}(X_i - \overline{X})^2\right] = n\sigma^2 - n\,\mathrm{Var}(\overline{X}).$$

Since

$$\mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n},$$

it follows that

$$\mathbb{E}\left[\sum_{i=1}^{n}(X_i - \overline{X})^2\right] = (n-1)\sigma^2.$$

Therefore,

$$\mathbb{E}[S^2] = \sigma^2.$$

□

Remark 2.7. This explains why the denominator of the sample variance is $n-1$ rather than n.

If instead we defined

$$\widetilde{S}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2,$$

then

$$\mathbb{E}[\widetilde{S}^2] = \frac{n-1}{n}\sigma^2,$$

so $\widetilde{S}^2$ would systematically underestimate σ^2.

2.3 Sampling from a Normal Population

The normal population is especially important because the exact sampling distributions of both $\overline{X}$ and S^2 can be obtained.

Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2).$$

Theorem 2.8 (Distribution of the Sample Mean)**.** *For a normal random sample,*

$$\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

Equivalently,

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1).$$

Proof. Since a linear combination of independent normal random variables is normal, $\overline{X}$ is normally distributed.

Furthermore,

$$\mathbb{E}[\overline{X}] = \mu$$

and

$$\mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n}.$$

Hence

$$\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

□

Theorem 2.9 (Distribution of the Sample Variance)**.** *If*

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2),$$

then

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

Theorem 2.10 (Independence of Sample Mean and Sample Variance)**.** *If*

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2),$$

then

$$\overline{X} \quad \textit{and} \quad S^2$$

are independent.

Remark 2.11. The independence of $\overline{X}$ and S^2 is a special property of the normal distribution. It is not true for an arbitrary population.

This independence is exactly what allows us to construct the Student t statistic when σ^2 is unknown.

Theorem 2.12 (Normal Sample Decomposition)**.** *For a normal random sample,*

$$\sum_{i=1}^{n} \frac{(X_i - \mu)^2}{\sigma^2} = \frac{(n-1)S^2}{\sigma^2} + \frac{n(\overline{X} - \mu)^2}{\sigma^2}.$$

The left-hand side satisfies

$$\sum_{i=1}^{n} \frac{(X_i - \mu)^2}{\sigma^2} \sim \chi^2_n,$$

while

$$\frac{n(\overline{X} - \mu)^2}{\sigma^2} \sim \chi^2_1.$$

Consequently,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

2.4 Student's t Distribution

Definition 2.13 (Student's t Distribution)**.** Let

$$Z \sim N(0,1)$$

and

$$U \sim \chi^2_\nu$$

be independent. Then

$$T = \frac{Z}{\sqrt{U/\nu}}$$

has a Student's t distribution with ν degrees of freedom, written

$$T \sim t_\nu.$$

The density of $T \sim t_\nu$ is

$$f_T(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)}\left(1+\frac{t^2}{\nu}\right)^{-(\nu+1)/2}, \qquad -\infty < t < \infty.$$

The distribution is symmetric about zero:

$$T \overset{\mathrm{d}}{=} -T.$$

Hence, if

$$t_{\alpha,\nu}$$

denotes the upper α quantile, then

$$\mathbb{P}(T > t_{\alpha,\nu}) = \alpha$$

and

$$\mathbb{P}(T < -t_{\alpha,\nu}) = \alpha.$$

Theorem 2.14 (The t Statistic for a Normal Sample)**.** *If*

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2),$$

then

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

Proof. We know that

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1),$$

and

$$U = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

Moreover, Z and U are independent. Hence

$$\frac{Z}{\sqrt{U/(n-1)}} \sim t_{n-1}.$$

But

$$\frac{Z}{\sqrt{U/(n-1)}} = \frac{\frac{\overline{X}-\mu}{\sigma/\sqrt{n}}}{S/\sigma} = \frac{\overline{X}-\mu}{S/\sqrt{n}}.$$

□

Remark 2.15. The t statistic replaces the unknown population standard deviation σ by the sample standard deviation S.

This additional uncertainty produces heavier tails than the standard normal distribution.

As the degrees of freedom increase,

$$t_\nu \xrightarrow{\text{d}} N(0,1) \qquad \text{as } \nu \to \infty.$$

2.5 F Distribution

Definition 2.16 (F Distribution)**.** Let

$$U \sim \chi^2_{\nu_1}, \qquad V \sim \chi^2_{\nu_2}$$

be independent. Then

$$F = \frac{U/\nu_1}{V/\nu_2}$$

has an F distribution with ν_1 and ν_2 degrees of freedom:

$$F \sim F_{\nu_1,\nu_2}.$$

The density is

$$f_F(x) = \frac{\Gamma\left(\frac{\nu_1+\nu_2}{2}\right)}{\Gamma\left(\frac{\nu_1}{2}\right)\Gamma\left(\frac{\nu_2}{2}\right)} \left(\frac{\nu_1}{\nu_2}\right)^{\nu_1/2} x^{\nu_1/2-1} \left(1+\frac{\nu_1}{\nu_2}x\right)^{-(\nu_1+\nu_2)/2}, \qquad x>0.$$

Proposition 2.17 (Reciprocal Property)**.** *If*

$$F \sim F_{\nu_1,\nu_2},$$

then

$$\frac{1}{F} \sim F_{\nu_2,\nu_1}.$$

Consequently, if $F_{\alpha;\nu_1,\nu_2}$ denotes the upper α quantile, then

$$F_{\alpha;\nu_1,\nu_2} = \frac{1}{F_{1-\alpha;\nu_2,\nu_1}}.$$

Proposition 2.18 (Relation Between t and F)**.** *If*

$$T \sim t_\nu,$$

then

$$T^2 \sim F_{1,\nu}.$$

Remark 2.19. The F distribution naturally appears when comparing two independent variance estimates.

This will later lead to confidence intervals and hypothesis tests for the ratio

$$\frac{\sigma_1^2}{\sigma_2^2}.$$

2.6 Order Statistics

Let

$$X_1, \ldots, X_n$$

be iid random variables from a continuous distribution with CDF F and density f.

After arranging the observations in increasing order,

$$X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)},$$

the random variables

$$X_{(1)}, \ldots, X_{(n)}$$

are called the *order statistics*.

Definition 2.20 (Order Statistics)**.** The k-th order statistic

$$X_{(k)}$$

is the k-th smallest value among

$$X_1, \ldots, X_n.$$

In particular,

$$X_{(1)} = \min\{X_1, \ldots, X_n\},$$

and

$$X_{(n)} = \max\{X_1, \ldots, X_n\}.$$

2.6.1 Distribution of the Maximum

For the maximum,

$$X_{(n)} = \max\{X_1, \ldots, X_n\},$$

we have

$$\begin{aligned} F_{X_{(n)}}(x) &= \mathbb{P}(X_{(n)} \leq x) \\ &= \mathbb{P}(X_1 \leq x, \ldots, X_n \leq x) \\ &= \prod_{i=1}^{n} \mathbb{P}(X_i \leq x) \\ &= [F(x)]^n. \end{aligned}$$

Therefore,

$$\boxed{F_{X_{(n)}}(x) = [F(x)]^n.}$$

Differentiating,

$$\boxed{f_{X_{(n)}}(x) = n[F(x)]^{n-1} f(x).}$$

2.6.2 Distribution of the Minimum

For

$$X_{(1)} = \min\{X_1, \ldots, X_n\},$$

it is convenient to use the survival probability:

$$\begin{aligned} \mathbb{P}(X_{(1)} > x) &= \mathbb{P}(X_1 > x, \ldots, X_n > x) \\ &= [1 - F(x)]^n. \end{aligned}$$

Hence

$$F_{X_{(1)}}(x) = 1 - [1 - F(x)]^n.$$

Therefore,

$$\boxed{f_{X_{(1)}}(x) = n[1 - F(x)]^{n-1} f(x).}$$

2.6.3 General Order Statistic

Theorem 2.21 (Density of the k-th Order Statistic)**.** *The density of*

$$X_{(k)}$$

is

$$\boxed{f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!}[F(x)]^{k-1}[1-F(x)]^{n-k}f(x).}$$

Proof. For $X_{(k)}$ to lie near x,

- exactly $k-1$ observations must lie below x;
- one observation must lie near x;
- the remaining $n-k$ observations must lie above x.

The corresponding probability is approximately

$$\frac{n!}{(k-1)!(n-k)!}[F(x)]^{k-1}f(x)\ \mathrm{d}x[1-F(x)]^{n-k}.$$

Dividing by $\mathrm{d}x$ gives the density. □

Example 2.22 (Uniform Order Statistics)**.** Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} U(0,1).$$

Then

$$F(x) = x, \qquad f(x) = 1, \qquad 0 < x < 1.$$

Therefore,

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!}x^{k-1}(1-x)^{n-k}.$$

Thus

$$\boxed{X_{(k)} \sim \mathrm{Beta}(k, n-k+1).}$$

In particular,

$$\mathbb{E}[X_{(k)}] = \frac{k}{n+1}.$$

Remark 2.23. Order statistics are important whenever inference depends on extreme observations, sample quantiles, medians, ranges, or the boundary of the support of a distribution.

2.7 Law of Large Numbers

One of the most important properties of the sample mean is that it becomes close to the population mean as the sample size grows.

Theorem 2.24 (Weak Law of Large Numbers)**.** *Let*

$$X_1, X_2, \ldots$$

be iid random variables with

$$\mathbb{E}[X_i] = \mu$$

and finite variance

$$\mathrm{Var}(X_i) = \sigma^2 < \infty.$$

Then

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$$

satisfies

$$\overline{X}_n \xrightarrow{\mathbb{P}} \mu.$$

Equivalently, for every $\varepsilon > 0$,

$$\mathbb{P}\left(|\overline{X}_n - \mu| > \varepsilon\right) \longrightarrow 0.$$

Proof. Since

$$\mathbb{E}[\overline{X}_n] = \mu$$

and

$$\mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n},$$

Chebyshev's inequality gives

$$\mathbb{P}\left(|\overline{X}_n - \mu| \geq \varepsilon\right) \leq \frac{\sigma^2}{n\varepsilon^2}.$$

As

$$n \to \infty,$$

the right-hand side tends to zero. □

Remark 2.25. The law of large numbers explains why the sample mean is a natural estimator of the population mean.

It says that with a sufficiently large sample,

$$\overline{X}_n$$

is likely to be close to μ.

2.8 Central Limit Theorem

The law of large numbers tells us that

$$\overline{X}_n \to \mu,$$

but it does not describe the shape of the fluctuations around μ.

The central limit theorem provides this information.

Theorem 2.26 (Central Limit Theorem)**.** *Let*

$$X_1, X_2, \ldots$$

be iid random variables satisfying

$$\mathbb{E}[X_i] = \mu, \qquad \mathrm{Var}(X_i) = \sigma^2 < \infty.$$

Then

$$\boxed{\frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma} \xrightarrow{\mathrm{d}} N(0,1).}$$

Equivalently,

$$\frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}} \xrightarrow{\mathrm{d}} N(0,1).$$

For sufficiently large n, we therefore use the approximation

$$\overline{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right).$$

Remark 2.27. There is an important distinction between the normal-sample theorem and the central limit theorem.

If

$$X_i \sim N(\mu, \sigma^2),$$

then

$$\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

exactly for every n.

For a general population with finite variance, the same normal distribution is only an approximation for sufficiently large n.

2.9 Normal Approximation to the Binomial Distribution

Suppose

$$X \sim \text{Binomial}(n, p).$$

Since

$$X = \sum_{i=1}^{n} X_i, \qquad X_i \overset{\text{iid}}{\sim} \text{Bernoulli}(p),$$

we have

$$\mathbb{E}[X] = np$$

and

$$\text{Var}(X) = np(1-p).$$

By the central limit theorem,

$$\boxed{\frac{X - np}{\sqrt{np(1-p)}} \approx N(0,1)}$$

for sufficiently large n.

Thus

$$X \approx N\Big(np,\, np(1-p)\Big).$$

Remark 2.28 (Continuity Correction). The binomial distribution is discrete whereas the normal distribution is continuous. A continuity correction can therefore improve the approximation.

For example,

$$\mathbb{P}(X \le k)$$

is approximated by

$$\mathbb{P}\left(Y \le k + \frac{1}{2}\right),$$

where

$$Y \sim N\Big(np, np(1-p)\Big).$$

Similarly,

$$\mathbb{P}(X \ge k)$$

is approximated by

$$\mathbb{P}\left(Y \ge k - \frac{1}{2}\right).$$

2.10 Poisson Approximation to the Binomial Distribution

Another useful approximation occurs when

$$n$$

is large and

$$p$$

is small.

Suppose

$$X_n \sim \text{Binomial}(n, p_n)$$

and

$$np_n \to \lambda.$$

Then

$$\boxed{X_n \xrightarrow{\text{d}} \text{Poisson}(\lambda).}$$

Thus, when n is large and p is small,

$$\text{Binomial}(n, p) \approx \text{Poisson}(np).$$

Remark 2.29. The normal and Poisson approximations apply in different regimes.

For a binomial random variable:

$$\boxed{\begin{array}{rcl} np \text{ and } n(1-p) \text{ large} & \implies & \text{normal approximation,} \\ n \text{ large and } p \text{ small} & \implies & \text{Poisson approximation.} \end{array}}$$

2.11 Summary of Important Sampling Distributions

For a normal random sample

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2),$$

the most important sampling distributions are

$$\boxed{\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)}$$

$$\boxed{\frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)}$$

$$\boxed{\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}}$$

$$\boxed{\overline{X} \perp S^2}$$

and therefore

$$\boxed{\frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.}$$

If

$$U \sim \chi^2_{\nu_1}, \qquad V \sim \chi^2_{\nu_2}, \qquad U \perp V,$$

then

$$\boxed{\frac{U/\nu_1}{V/\nu_2} \sim F_{\nu_1,\nu_2}.}$$

These formulas will be used repeatedly in the construction of confidence intervals and hypothesis tests.

Chapter 3

Point Estimation

3 Point Estimation

3.1 Basic Concepts of Point Estimation

Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x; \theta),$$

where $\theta \in \Theta$ is an unknown parameter.

The objective of point estimation is to use the observed sample to construct a single numerical approximation to θ, or more generally to some function of θ.

Definition 3.1 (Estimator)**.** An *estimator* of a parameter function $\tau(\theta)$ is a statistic

$$\widehat{\tau} = T(X_1, \ldots, X_n)$$

used to estimate $\tau(\theta)$.

Before observing the sample,

$$\widehat{\tau}$$

is a random variable.

After observing

$$X_1 = x_1, \ldots, X_n = x_n,$$

the numerical value

$$\widehat{\tau}(x_1, \ldots, x_n)$$

is called an *estimate*.

Remark 3.2. An estimator must be a statistic and therefore cannot depend on an unknown parameter.

For example, if μ is unknown, then

$$\overline{X}$$

is a valid estimator of μ, whereas

$$\overline{X} + \mu$$

is not an estimator because it still contains the unknown parameter μ.

3.2 Method of Moments

The method of moments estimates unknown parameters by matching theoretical moments with sample moments.

Suppose the model contains k unknown parameters

$$\theta_1, \ldots, \theta_k.$$

The first k population moments are

$$\mu_j' = \mathbb{E}_\theta[X^j], \qquad j = 1, \ldots, k.$$

The corresponding sample moments are

$$M_j' = \frac{1}{n} \sum_{i=1}^{n} X_i^j.$$

Definition 3.3 (Method of Moments Estimator)**.** The method of moments estimators

$$\widehat{\theta}_1, \ldots, \widehat{\theta}_k$$

are obtained by solving

$$\mathbb{E}_\theta[X] = \frac{1}{n}\sum_{i=1}^{n} X_i,$$

$$\mathbb{E}_\theta[X^2] = \frac{1}{n}\sum_{i=1}^{n} X_i^2,$$

$$\vdots$$

$$\mathbb{E}_\theta[X^k] = \frac{1}{n}\sum_{i=1}^{n} X_i^k.$$

In compact form,

$$\boxed{\text{population moments} = \text{sample moments.}}$$

Example 3.4 (Bernoulli Distribution)**.** Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p).$$

Since

$$\mathbb{E}[X] = p,$$

the method of moments equation is

$$p = \overline{X}.$$

Therefore,

$$\boxed{\widehat{p}_{\text{MOM}} = \overline{X}.}$$

Example 3.5 (Exponential Distribution)**.** Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Exponential}(\lambda),$$

with rate parameter λ.

Since

$$\mathbb{E}[X] = \frac{1}{\lambda},$$

the moment equation is

$$\frac{1}{\lambda} = \overline{X}.$$

Hence

$$\boxed{\widehat{\lambda}_{\text{MOM}} = \frac{1}{\overline{X}}.}$$

Example 3.6 (Gamma Distribution)**.** Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Gamma}(\alpha, \lambda),$$

using the shape–rate parameterization.

Then

$$\mathbb{E}[X] = \frac{\alpha}{\lambda},$$

and

$$\mathrm{Var}(X) = \frac{\alpha}{\lambda^2}.$$

Equivalently,

$$\mathbb{E}[X^2] = \frac{\alpha(\alpha+1)}{\lambda^2}.$$

The method of moments equations are

$$\frac{\alpha}{\lambda} = \overline{X},$$

and

$$\frac{\alpha(\alpha+1)}{\lambda^2} = \frac{1}{n}\sum_{i=1}^{n} X_i^2.$$

Solving these equations gives estimators of α and λ.

Remark 3.7. The method of moments is usually computationally simple, but it does not necessarily produce the statistically most efficient estimator.

Its main advantage is that it provides an estimator without requiring optimization of the likelihood function.

3.3 Maximum Likelihood Estimation

Maximum likelihood estimation chooses the parameter value under which the observed data are most plausible.

Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x;\theta).$$

Definition 3.8 (Likelihood Function)**.** For observed values

$$x_1, \ldots, x_n,$$

the likelihood function is

$$L(\theta) = L(\theta; x_1, \ldots, x_n) = \prod_{i=1}^{n} f(x_i;\theta).$$

When the sample is random, we may write

$$L(\theta; X) = \prod_{i=1}^{n} f(X_i;\theta).$$

Definition 3.9 (Maximum Likelihood Estimator)**.** A maximum likelihood estimator is a value

$$\widehat{\theta}_{\text{MLE}}$$

satisfying

$$L(\widehat{\theta}_{\text{MLE}}) = \sup_{\theta\in\Theta} L(\theta).$$

Because products are often difficult to differentiate, we usually work with the log-likelihood.

Definition 3.10 (Log-Likelihood)**.** The log-likelihood function is

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log f(X_i; \theta).$$

Since $\log x$ is strictly increasing,

$$\arg\max_{\theta} L(\theta) = \arg\max_{\theta} \ell(\theta).$$

If the maximum occurs at an interior point and the likelihood is differentiable, a candidate MLE satisfies

$$\frac{\partial}{\partial \theta} \ell(\theta) = 0.$$

For a vector parameter

$$\theta = (\theta_1, \ldots, \theta_k),$$

we solve

$$\frac{\partial \ell}{\partial \theta_j} = 0, \qquad j = 1, \ldots, k.$$

Remark 3.11. Solving

$$\ell'(\theta) = 0$$

only gives candidate extrema.

One must still check that the solution belongs to the parameter space and actually maximizes the likelihood. If the maximum occurs on the boundary of the parameter space, the likelihood equation may have no interior solution.

3.3.1 Example: Bernoulli Model

Example 3.12 (MLE for a Bernoulli Parameter)**.** Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p).$$

The likelihood function is

$$L(p) = \prod_{i=1}^{n} p^{X_i} (1-p)^{1-X_i}.$$

Therefore,

$$L(p) = p^{\sum X_i} (1-p)^{n - \sum X_i}.$$

The log-likelihood is

$$\ell(p) = \left(\sum_{i=1}^{n} X_i \right) \log p + \left(n - \sum_{i=1}^{n} X_i \right) \log(1-p).$$

Differentiating,

$$\ell'(p) = \frac{\sum X_i}{p} - \frac{n - \sum X_i}{1-p}.$$

Setting

$$\ell'(p) = 0$$

gives

$$\left(\sum X_i \right) (1-p) = p \left(n - \sum X_i \right).$$

Hence

$$np = \sum_{i=1}^{n} X_i,$$

so

$$\boxed{\widehat{p}_{\text{MLE}} = \overline{X}.}$$

3.3.2 Example: Poisson Model

Example 3.13 (MLE for a Poisson Parameter)**.** Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Poisson}(\lambda).$$

Then

$$L(\lambda) = \prod_{i=1}^{n} \frac{e^{-\lambda}\lambda^{X_i}}{X_i!}.$$

Thus

$$L(\lambda) = e^{-n\lambda}\lambda^{\sum X_i} \prod_{i=1}^{n} \frac{1}{X_i!}.$$

The log-likelihood is

$$\ell(\lambda) = -n\lambda + \left(\sum_{i=1}^{n} X_i\right) \log \lambda - \sum_{i=1}^{n} \log(X_i!).$$

Hence

$$\ell'(\lambda) = -n + \frac{\sum X_i}{\lambda}.$$

Setting this equal to zero gives

$$\boxed{\widehat{\lambda}_{\text{MLE}} = \overline{X}.}$$

3.3.3 Example: Normal Model

Example 3.14 (MLE for Normal Mean and Variance)**.** Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2),$$

where both μ and σ^2 are unknown.

The likelihood function is

$$L(\mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2\right\}.$$

Therefore,

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2.$$

Differentiating with respect to μ gives

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \mu).$$

Hence

$$\widehat{\mu}_{\text{MLE}} = \overline{X}.$$

Differentiating with respect to σ^2 gives

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(X_i - \mu)^2.$$

Substituting

$$\widehat{\mu} = \overline{X}$$

gives

$$\boxed{\widehat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2.}$$

Remark 3.15. The MLE of σ^2 is

$$\widehat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2,$$

whereas the unbiased sample variance is

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2.$$

Thus

$$\widehat{\sigma}^2_{\text{MLE}} \neq S^2.$$

Maximum likelihood estimation does not automatically produce an unbiased estimator.

3.4 Invariance Property of the MLE

Theorem 3.16 (Invariance Property)**.** *Suppose*

$$\widehat{\theta}_{\text{MLE}}$$

is the MLE of θ*, and*

$$\tau = g(\theta).$$

Then, under the usual interpretation of the induced parameterization, the MLE of τ *is*

$$\boxed{\widehat{\tau}_{\text{MLE}} = g(\widehat{\theta}_{\text{MLE}}).}$$

Example 3.17. If

$$\widehat{\sigma^2}_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2,$$

then the MLE of σ is

$$\widehat{\sigma}_{\text{MLE}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2}.$$

3.5 Bayesian Point Estimation

The frequentist approach treats θ as an unknown but fixed parameter. The Bayesian approach instead models θ itself as a random variable.

Suppose the prior density of θ is

$$\pi(\theta).$$

Given observed data

$$x = (x_1, \ldots, x_n),$$

the likelihood is

$$L(\theta; x).$$

Theorem 3.18 (Bayes' Formula for the Posterior)**.** *The posterior density of* θ *is*

$$\boxed{\pi(\theta \mid x) = \frac{L(\theta; x)\pi(\theta)}{\int_\Theta L(u; x)\pi(u)\ \mathrm{d}u}.}$$

Equivalently,

$$\pi(\theta \mid x) \propto L(\theta; x)\pi(\theta).$$

Common Bayesian point estimators include:

$$\widehat{\theta}_{\text{posterior mean}} = \mathbb{E}[\theta \mid X],$$

$$\widehat{\theta}_{\text{MAP}} = \arg\max_{\theta} \pi(\theta \mid X),$$

where MAP stands for *maximum a posteriori*.

Remark 3.19. Maximum likelihood uses only the likelihood:

$$L(\theta; X).$$

Bayesian estimation combines the likelihood with prior information:

$$L(\theta; X)\pi(\theta).$$

Thus MLE and Bayesian estimation answer related but conceptually different questions.

3.6 Bias and Unbiased Estimation

Once an estimator has been constructed, we need criteria for comparing its quality.

Definition 3.20 (Bias)**.** The bias of an estimator $\widehat{\theta}$ is

$$\text{Bias}_\theta(\widehat{\theta}) = \mathbb{E}_\theta[\widehat{\theta}] - \theta.$$

Definition 3.21 (Unbiased Estimator)**.** An estimator $\widehat{\theta}$ is unbiased for θ if

$$\boxed{\mathbb{E}_\theta[\widehat{\theta}] = \theta}$$

for every

$$\theta \in \Theta.$$

More generally, T is unbiased for $\tau(\theta)$ if

$$\mathbb{E}_\theta[T] = \tau(\theta).$$

Example 3.22 (Sample Mean)**.** If

$$X_1, \ldots, X_n$$

are iid with

$$\mathbb{E}[X_i] = \mu,$$

then

$$\mathbb{E}[\overline{X}] = \mu.$$

Hence

$$\overline{X}$$

is an unbiased estimator of μ.

Example 3.23 (Bias of the Normal MLE of Variance)**.** For a normal random sample,

$$\widehat{\sigma}^2_{\mathrm{MLE}} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2.$$

Since

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$$

satisfies

$$\mathbb{E}[S^2] = \sigma^2,$$

we obtain

$$\mathbb{E}\left[\widehat{\sigma}^2_{\mathrm{MLE}}\right] = \frac{n-1}{n}\sigma^2.$$

Therefore,

$$\mathrm{Bias}\left(\widehat{\sigma}^2_{\mathrm{MLE}}\right) = -\frac{\sigma^2}{n}.$$

3.7 Mean Squared Error

Definition 3.24 (Mean Squared Error)**.** The mean squared error of an estimator $\widehat{\theta}$ is

$$\mathrm{MSE}_\theta(\widehat{\theta}) = \mathbb{E}_\theta\left[(\widehat{\theta} - \theta)^2\right].$$

Theorem 3.25 (Bias–Variance Decomposition)**.** *For any estimator $\widehat{\theta}$,*

$$\boxed{\mathrm{MSE}(\widehat{\theta}) = \mathrm{Var}(\widehat{\theta}) + \mathrm{Bias}(\widehat{\theta})^2.}$$

Proof. Write

$$\widehat{\theta} - \theta = \left(\widehat{\theta} - \mathbb{E}[\widehat{\theta}]\right) + \left(\mathbb{E}[\widehat{\theta}] - \theta\right).$$

Squaring and taking expectations gives

$$\mathrm{MSE}(\widehat{\theta}) = \mathrm{Var}(\widehat{\theta}) + \left(\mathbb{E}[\widehat{\theta}] - \theta\right)^2,$$

because

$$\mathbb{E}\left[\widehat{\theta} - \mathbb{E}[\widehat{\theta}]\right] = 0.$$

□

If $\widehat{\theta}$ is unbiased, then

$$\mathrm{MSE}(\widehat{\theta}) = \mathrm{Var}(\widehat{\theta}).$$

Remark 3.26. Unbiasedness alone does not guarantee that an estimator is good.

A slightly biased estimator may have much smaller variance and hence smaller MSE than an unbiased estimator.

Thus MSE measures the combined effect of both bias and variability.

3.8 Consistency

Definition 3.27 (Consistent Estimator)**.** An estimator sequence

$$\widehat{\theta}_n$$

is consistent for θ if

$$\boxed{\widehat{\theta}_n \xrightarrow{\mathbb{P}} \theta.}$$

Equivalently, for every $\varepsilon > 0$,

$$\mathbb{P}_\theta\left(|\widehat{\theta}_n - \theta| > \varepsilon\right) \longrightarrow 0.$$

Theorem 3.28 (A Sufficient Condition for Consistency)**.** *If*

$$\mathrm{Bias}(\widehat{\theta}_n) \longrightarrow 0$$

and

$$\mathrm{Var}(\widehat{\theta}_n) \longrightarrow 0,$$

then

$$\widehat{\theta}_n \xrightarrow{\mathbb{P}} \theta.$$

Example 3.29. Since

$$\mathbb{E}[\overline{X}_n] = \mu$$

and

$$\mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n} \longrightarrow 0,$$

the sample mean is consistent:

$$\overline{X}_n \xrightarrow{\mathbb{P}} \mu.$$

Remark 3.30. Unbiasedness and consistency are different properties.

Unbiasedness concerns expectation at a fixed sample size:

$$\mathbb{E}[\widehat{\theta}_n] = \theta.$$

Consistency concerns limiting behavior:

$$\widehat{\theta}_n \xrightarrow{\mathbb{P}} \theta.$$

Neither concept should be confused with the other.

3.9 Score Function and Fisher Information

The likelihood not only produces estimators but also measures how much information the sample contains about the parameter.

Definition 3.31 (Score Function)**.** For a scalar parameter θ, the score function is

$$\boxed{S(\theta; X) = \frac{\partial}{\partial\theta} \log L(\theta; X).}$$

For an iid sample,

$$S(\theta; X) = \sum_{i=1}^{n} \frac{\partial}{\partial \theta} \log f(X_i; \theta).$$

Under appropriate regularity conditions,

$$\mathbb{E}_\theta[S(\theta; X)] = 0.$$

Proof. For one observation,

$$\mathbb{E}_\theta \left[\frac{\partial}{\partial \theta} \log f(X; \theta) \right] = \int \frac{\partial}{\partial \theta} \log f(x; \theta) f(x; \theta) \, \mathrm{d}x.$$

Since

$$\frac{\partial}{\partial \theta} \log f(x; \theta) = \frac{\frac{\partial}{\partial \theta} f(x; \theta)}{f(x; \theta)},$$

we obtain

$$\mathbb{E}_\theta[S(\theta; X)] = \int \frac{\partial}{\partial \theta} f(x; \theta) \, \mathrm{d}x.$$

Under conditions allowing differentiation under the integral sign,

$$= \frac{\partial}{\partial \theta} \int f(x; \theta) \, \mathrm{d}x = \frac{\partial}{\partial \theta}(1) = 0.$$

□

Definition 3.32 (Fisher Information)**.** For one observation, the Fisher information is

$$I_1(\theta) = \mathbb{E}_\theta \left[\left(\frac{\partial}{\partial \theta} \log f(X; \theta) \right)^2 \right].$$

For a sample,

$$I_n(\theta) = \mathbb{E}_\theta \left[S(\theta; X)^2 \right].$$

Under suitable regularity conditions,

$$\boxed{I_1(\theta) = -\mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta^2} \log f(X; \theta) \right].}$$

For iid observations,

$$\boxed{I_n(\theta) = n I_1(\theta).}$$

Remark 3.33. Fisher information measures how strongly the probability distribution changes when θ changes.

Large Fisher information means that nearby parameter values generate more distinguishable distributions, so the parameter can potentially be estimated more accurately.

Example 3.34 (Fisher Information for Bernoulli Data)**.** Suppose

$$X \sim \text{Bernoulli}(p).$$

Then

$$\log f(X; p) = X \log p + (1 - X) \log(1 - p).$$

Hence

$$\frac{\partial}{\partial p}\log f(X;p) = \frac{X}{p} - \frac{1-X}{1-p}.$$

The Fisher information for one observation is

$$I_1(p) = \frac{1}{p(1-p)}.$$

Therefore, for n iid observations,

$$\boxed{I_n(p) = \frac{n}{p(1-p)}.}$$

3.10 Cramér–Rao Inequality

The Fisher information determines a fundamental lower bound on the variance of unbiased estimators.

Theorem 3.35 (Cramér–Rao Lower Bound)**.** *Suppose $T(X)$ is an unbiased estimator of θ, and suitable regularity conditions hold. Then*

$$\boxed{\mathrm{Var}_\theta(T) \geq \frac{1}{I_n(\theta)}.}$$

More generally, suppose T is unbiased for a differentiable function

$$\tau(\theta).$$

Then

$$\boxed{\mathrm{Var}_\theta(T) \geq \frac{[\tau'(\theta)]^2}{I_n(\theta)}.}$$

Proof. Let

$$S(\theta;X) = \frac{\partial}{\partial\theta}\log L(\theta;X)$$

be the score function.

Since

$$\mathbb{E}_\theta[T] = \tau(\theta),$$

differentiate with respect to θ:

$$\tau'(\theta) = \frac{\partial}{\partial\theta}\mathbb{E}_\theta[T].$$

Under regularity conditions,

$$\tau'(\theta) = \mathbb{E}_\theta\left[TS(\theta;X)\right].$$

Since

$$\mathbb{E}[S(\theta;X)] = 0,$$

we obtain

$$\tau'(\theta) = \mathrm{Cov}\left(T, S(\theta;X)\right).$$

By Cauchy–Schwarz,

$$[\tau'(\theta)]^2 \leq \mathrm{Var}(T)\,\mathrm{Var}(S).$$

But

$$\mathrm{Var}(S) = I_n(\theta).$$

Therefore,

$$\mathrm{Var}(T) \geq \frac{[\tau'(\theta)]^2}{I_n(\theta)}.$$

□

Definition 3.36 (Efficient Estimator)**.** An unbiased estimator T is called *efficient* if it attains the Cramér–Rao lower bound:

$$\mathrm{Var}(T) = \frac{[\tau'(\theta)]^2}{I_n(\theta)}.$$

Example 3.37 (Efficiency of the Bernoulli Sample Mean)**.** Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p).$$

The estimator

$$\overline{X}$$

is unbiased:

$$\mathbb{E}[\overline{X}] = p.$$

Its variance is

$$\mathrm{Var}(\overline{X}) = \frac{p(1-p)}{n}.$$

Since

$$I_n(p) = \frac{n}{p(1-p)},$$

the Cramér–Rao lower bound is

$$\frac{1}{I_n(p)} = \frac{p(1-p)}{n}.$$

Thus

$$\mathrm{Var}(\overline{X}) = \frac{1}{I_n(p)},$$

so $\overline{X}$ is efficient.

3.11 Uniformly Minimum Variance Unbiased Estimation

Definition 3.38 (UMVUE)**.** Let T be an unbiased estimator of $\tau(\theta)$.

Then T is called a *uniformly minimum variance unbiased estimator* (UMVUE) if

$$\mathrm{Var}_\theta(T) \leq \mathrm{Var}_\theta(U)$$

for every $\theta \in \Theta$ and every other unbiased estimator U of $\tau(\theta)$.

Remark 3.39. The word *uniformly* means that the variance comparison holds for every parameter value

$$\theta \in \Theta,$$

not merely at one particular value of θ.

An efficient unbiased estimator is automatically UMVUE whenever the Cramér–Rao lower bound applies to the entire class of unbiased estimators.

However, an UMVUE need not necessarily attain the Cramér–Rao bound.

Remark 3.40. These concepts should therefore be distinguished:

$$\boxed{\text{efficient} \Longrightarrow \text{UMVUE}}$$

when the relevant Cramér–Rao bound applies, but the converse need not hold.

3.12 Sufficient Statistics

A statistic is useful not only because it reduces the size of the data, but because it may preserve all information about the unknown parameter.

The original sample

$$X = (X_1, \ldots, X_n)$$

may be regarded as *raw data*, while

$$T(X)$$

is a reduced representation of the data.

Definition 3.41 (Sufficient Statistic)**.** A statistic

$$T = T(X)$$

is sufficient for θ if the conditional distribution of the full sample X, given

$$T(X) = t,$$

does not depend on θ.

In other words,

$$\boxed{\mathcal{L}\left(X \mid T(X) = t\right) \text{ does not depend on } \theta.}$$

Remark 3.42. Once the value of a sufficient statistic is known, the remaining details of the sample contain no additional information about θ.

Thus sufficiency formalizes the idea of lossless data reduction:

$$\boxed{\text{raw data} \longrightarrow \text{sufficient statistic}}$$

without losing information about the parameter.

Example 3.43 (Poisson Sample)**.** Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Poisson}(\lambda).$$

Then

$$T(X) = \sum_{i=1}^{n} X_i$$

is sufficient for λ.

3.13 Factorization Theorem

Checking the conditional-distribution definition of sufficiency directly can be difficult. The factorization theorem provides a much more convenient criterion.

Theorem 3.44 (Neyman–Fisher Factorization Theorem)**.** *A statistic*

$$T = T(X)$$

is sufficient for θ if and only if the joint density or joint mass function of the sample can be written as

$$\boxed{f(x_1, \ldots, x_n; \theta) = g(T(x); \theta) h(x),}$$

where

$$h(x)$$

does not depend on θ, and

$$g(T(x); \theta)$$

depends on the sample only through $T(x)$.

More generally, a collection

$$T_1(X), \ldots, T_k(X)$$

is jointly sufficient if

$$f(x; \theta) = g\left(T_1(x), \ldots, T_k(x); \theta\right) h(x).$$

Example 3.45 (Poisson Sufficiency)**.** Let

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Poisson}(\lambda).$$

The joint mass function is

$$\begin{aligned} f(x_1, \ldots, x_n; \lambda) &= \prod_{i=1}^{n} \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} \\ &= e^{-n\lambda} \lambda^{\sum_{i=1}^{n} x_i} \prod_{i=1}^{n} \frac{1}{x_i!}. \end{aligned}$$

Write

$$g(T; \lambda) = e^{-n\lambda} \lambda^T, \qquad T = \sum_{i=1}^{n} x_i,$$

and

$$h(x) = \prod_{i=1}^{n} \frac{1}{x_i!}.$$

Since h does not depend on λ, the factorization theorem implies

$$\boxed{T = \sum_{i=1}^{n} X_i}$$

is sufficient for λ.

Example 3.46 (Normal Distribution with Unknown Mean and Variance)**.** Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2),$$

where both parameters are unknown.

The joint density is

$$f(x; \mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2 \right\}.$$

Using

$$\sum_{i=1}^{n} (x_i - \mu)^2 = \sum_{i=1}^{n} x_i^2 - 2\mu \sum_{i=1}^{n} x_i + n\mu^2,$$

the density depends on the sample only through

$$\sum_{i=1}^{n} X_i$$

and

$$\sum_{i=1}^{n} X_i^2.$$

Therefore,

$$\boxed{\left(\sum_{i=1}^{n} X_i, \sum_{i=1}^{n} X_i^2 \right)}$$

is jointly sufficient for

$$(\mu, \sigma^2).$$

Remark 3.47. Equivalent one-to-one transformations of a sufficient statistic remain sufficient.

For the normal model, instead of

$$\left(\sum X_i, \sum X_i^2 \right),$$

one may use

$$(\overline{X}, S^2).$$

3.14 Rao–Blackwell Improvement

Sufficiency becomes especially useful when constructing good unbiased estimators.

Theorem 3.48 (Rao–Blackwell Theorem)**.** *Let U be an unbiased estimator of $\tau(\theta)$ and let T be sufficient for θ.*

Define

$$U^* = \mathbb{E}[U \mid T].$$

Then U^* is also unbiased for $\tau(\theta)$ and

$$\boxed{\operatorname{Var}(U^*) \leq \operatorname{Var}(U).}$$

Proof. By the law of total expectation,

$$\mathbb{E}[U^*] = \mathbb{E}\left[\mathbb{E}[U \mid T]\right] = \mathbb{E}[U] = \tau(\theta).$$

Therefore U^* is unbiased.
By the law of total variance,

$$\mathrm{Var}(U) = \mathbb{E}\left[\mathrm{Var}(U \mid T)\right] + \mathrm{Var}\left(\mathbb{E}[U \mid T]\right).$$

Thus

$$\mathrm{Var}(U) \geq \mathrm{Var}(U^*).$$

□

Remark 3.49. Rao–Blackwellization says that if a sufficient statistic is available, then an unbiased estimator can usually be improved by conditioning on that statistic:

$$U \quad \longrightarrow \quad \mathbb{E}[U \mid T].$$

The resulting estimator depends only on the sufficient statistic and cannot have larger variance.

3.15 Complete Statistics

Sufficiency says that a statistic contains all information about the parameter. Completeness is a different property that prevents distinct unbiased functions of that statistic from estimating zero.

Definition 3.50 (Complete Statistic)**.** A statistic T is complete for the family

$$\{P_\theta : \theta \in \Theta\}$$

if

$$\mathbb{E}_\theta[g(T)] = 0 \qquad \text{for every } \theta \in \Theta$$

implies

$$\boxed{\mathbb{P}_\theta(g(T) = 0) = 1}$$

for every $\theta \in \Theta$.

Equivalently,

$$\mathbb{E}_\theta[g(T)] = 0 \quad \forall\theta$$

implies

$$g(T) = 0 \quad \text{almost surely.}$$

Remark 3.51. Sufficiency and completeness describe different properties.

- Sufficiency concerns how much information about θ is retained.
- Completeness concerns uniqueness of unbiased functions of the statistic.

Thus neither concept should simply be interpreted as "stronger" than the other.

3.16 Lehmann–Scheffé Theorem

The combination of sufficiency and completeness gives one of the strongest methods for identifying an optimal unbiased estimator.

Theorem 3.52 (Lehmann–Scheffé Theorem)**.** *Suppose T is a complete sufficient statistic for θ.*

If

$$g(T)$$

is unbiased for

$$\tau(\theta),$$

then

$$\boxed{g(T) \text{ is the unique UMVUE of } \tau(\theta).}$$

Proof. Let U be any unbiased estimator of $\tau(\theta)$.

By the Rao–Blackwell theorem,

$$U^* = \mathbb{E}[U \mid T]$$

is an unbiased function of T satisfying

$$\operatorname{Var}(U^*) \leq \operatorname{Var}(U).$$

Since both U^* and $g(T)$ are unbiased,

$$\mathbb{E}_\theta\left[U^* - g(T)\right] = 0$$

for every θ.

Because T is complete,

$$U^* = g(T)$$

almost surely.

Therefore,

$$\operatorname{Var}(g(T)) \leq \operatorname{Var}(U)$$

for every unbiased estimator U.

Hence $g(T)$ is UMVUE. Completeness also gives uniqueness. □

Remark 3.53. The Lehmann–Scheffé theorem provides a powerful recipe for finding an UMVUE:

$$\boxed{\begin{array}{c}\text{Find a complete sufficient statistic } T \\ \Downarrow \\ \text{Find an unbiased function } g(T) \\ \Downarrow \\ g(T) \text{ is the unique UMVUE.}\end{array}}$$

3.17 Example: Poisson UMVUE

Example 3.54 (UMVUE in a Poisson Model)**.** Suppose

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Poisson}(\lambda).$$

We know that

$$T = \sum_{i=1}^{n} X_i$$

is sufficient for λ.

Moreover,

$$T \sim \text{Poisson}(n\lambda),$$

and this statistic is complete for λ.

Since

$$\mathbb{E}[T] = n\lambda,$$

we have

$$\mathbb{E}\left[\frac{T}{n}\right] = \lambda.$$

Therefore,

$$\boxed{\widehat{\lambda} = \frac{T}{n} = \overline{X}}$$

is an unbiased function of a complete sufficient statistic.

By the Lehmann–Scheffé theorem,

$$\boxed{\overline{X} \text{ is the unique UMVUE of } \lambda.}$$

3.18 Relationships Among Estimation Concepts

Several notions of estimator quality appear in this chapter, and they should not be confused.

Unbiased	$\iff$	$\mathbb{E}[\widehat{\theta}] = \theta$,
Consistent	$\iff$	$\widehat{\theta}_n \xrightarrow{\mathbb{P}} \theta$,
Efficient	$\iff$	attains the Cramér–Rao bound,
UMVUE	$\iff$	minimum variance among all unbiased estimators,
Sufficient	$\iff$	retains all sample information about θ,
Complete	$\iff$	guarantees uniqueness of unbiased functions.

Remark 3.55. The main logical connections are

$$\boxed{\text{complete} + \text{sufficient} + \text{unbiased function} \Longrightarrow \text{unique UMVUE},}$$

by the Lehmann–Scheffé theorem, and

$$\boxed{\text{sufficient statistic} + \text{unbiased estimator} \Longrightarrow \text{Rao–Blackwell improvement.}}$$

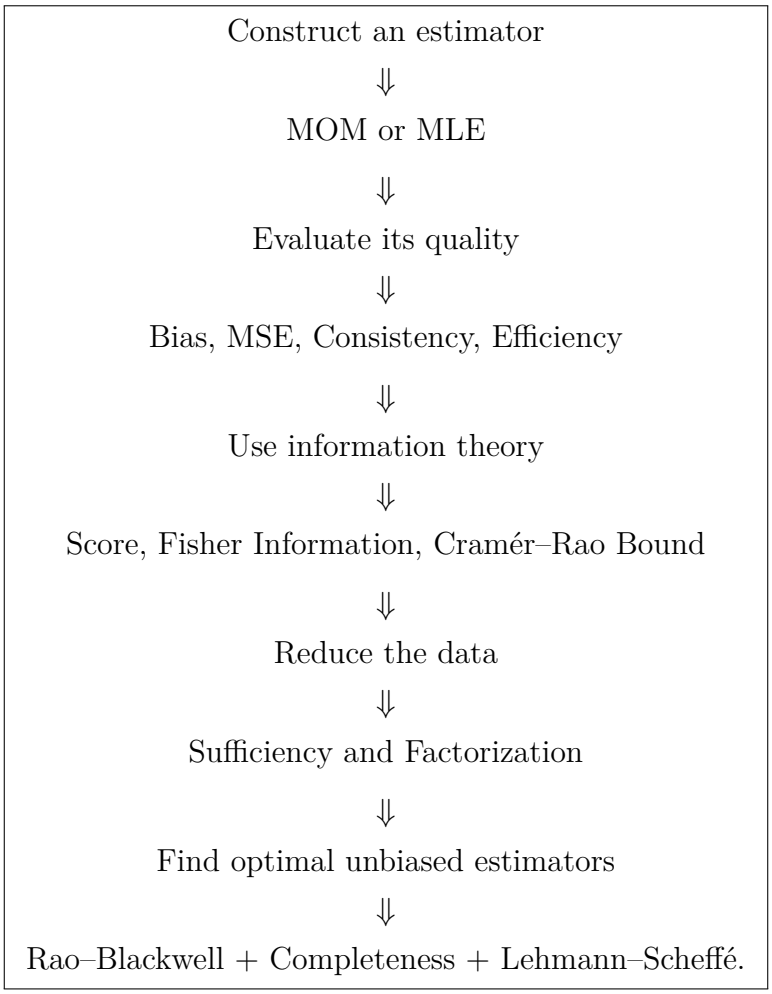
Also, whenever the Cramér–Rao bound applies and an unbiased estimator attains it,

$$\boxed{\text{efficient} \Longrightarrow \text{UMVUE.}}$$

However, unbiasedness alone does not imply efficiency, consistency, or sufficiency.

3.19 Summary of Point Estimation

The main structure of point estimation can be summarized as



Chapter 4

Confidence Interval Estimation

4 Confidence Interval Estimation

4.1 Introduction to Confidence Intervals

A point estimator gives a single value for an unknown parameter, but it does not directly describe the uncertainty caused by random sampling.

Confidence interval estimation provides a range of plausible values for the unknown parameter.

Definition 4.1 (Confidence Interval)**.** Let θ be an unknown parameter. Suppose

$$L = L(X_1, \ldots, X_n)$$

and

$$U = U(X_1, \ldots, X_n)$$

are statistics satisfying

$$\mathbb{P}_\theta\left(L \leq \theta \leq U\right) = 1 - \alpha.$$

Then

$$[L, U]$$

is called a $(1-\alpha)$ confidence interval for θ.

The quantity

$$1 - \alpha$$

is called the *confidence coefficient* or *confidence level*.

Common choices are

$$1 - \alpha = 0.90, \qquad 0.95, \qquad 0.99.$$

Remark 4.2. Before observing the sample,

$$L(X_1, \ldots, X_n) \quad \text{and} \quad U(X_1, \ldots, X_n)$$

are random variables, so the interval itself is random.

After the data have been observed, the endpoints become fixed numbers.

Thus a 95% confidence interval does *not* mean that the fixed parameter has probability 0.95 of lying inside one particular realized interval. Rather, the interval-construction procedure captures the true parameter in 95% of repeated samples.

4.2 Pivotal Quantities

The principal method for constructing confidence intervals in this course is based on pivotal quantities.

Definition 4.3 (Pivotal Quantity)**.** A function

$$Q = Q(X_1, \ldots, X_n; \theta)$$

is called a *pivotal quantity* if its probability distribution does not depend on any unknown parameter.

Suppose

$$Q(X;\theta)$$

has a known distribution and constants a and b satisfy

$$\mathbb{P}\left(a \leq Q(X;\theta) \leq b\right) = 1-\alpha.$$

If the inequality can be solved for θ, then we obtain

$$\mathbb{P}\left(L(X) \leq \theta \leq U(X)\right) = 1-\alpha,$$

which gives a $(1-\alpha)$ confidence interval.

Remark 4.4. The general construction is therefore

Find a pivot $\longrightarrow$ write a probability statement $\longrightarrow$ solve the inequality for θ.

4.3 Quantiles

Confidence intervals are usually written in terms of distribution quantiles.

For a standard normal random variable

$$Z \sim N(0,1),$$

let

$$z_\alpha$$

denote the upper α quantile:

$$\mathbb{P}(Z > z_\alpha) = \alpha.$$

Equivalently,

$$\mathbb{P}(Z \leq z_\alpha) = 1-\alpha.$$

By symmetry,

$$\mathbb{P}\left(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}\right) = 1-\alpha.$$

Similarly, if

$$T \sim t_\nu,$$

we use

$$t_{\alpha,\nu}$$

for the upper α quantile.

For

$$U \sim \chi^2_\nu,$$

we write

$$\chi^2_{\alpha,\nu}$$

for the upper α quantile:

$$\mathbb{P}\left(U > \chi^2_{\alpha,\nu}\right) = \alpha.$$

For

$$F \sim F_{\nu_1,\nu_2},$$

we write

$$F_{\alpha;\nu_1,\nu_2}$$

for the corresponding upper-tail quantile.

4.4 Confidence Interval for a Normal Mean

Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2).$$

We want to construct a confidence interval for μ.

4.4.1 Variance Known

Assume first that σ^2 is known.

Since

$$\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right),$$

the quantity

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}}$$

satisfies

$$Z \sim N(0, 1).$$

Therefore,

$$\mathbb{P}\left(-z_{\alpha/2} \leq \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right) = 1 - \alpha.$$

Solving for μ gives

$$\boxed{\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}.}$$

Hence the $(1 - \alpha)$ confidence interval is

$$\boxed{\left[\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right].}$$

Remark 4.5. The interval has the general form

$$\boxed{\text{estimate} \pm \text{critical value} \times \text{standard error.}}$$

Here,

$$\text{SE}(\overline{X}) = \frac{\sigma}{\sqrt{n}}.$$

4.4.2 Variance Unknown

If σ^2 is unknown, replace σ by the sample standard deviation S.

For a normal sample,

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

Thus

$$\mathbb{P}\left(-t_{\alpha/2,n-1} \leq \frac{\overline{X} - \mu}{S/\sqrt{n}} \leq t_{\alpha/2,n-1}\right) = 1 - \alpha.$$

Therefore,

$$\boxed{\left[\overline{X} - t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}, \overline{X} + t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}\right]}$$

is a $(1 - \alpha)$ confidence interval for μ.

Remark 4.6. The distinction between the two cases is fundamental:

$$\boxed{\begin{array}{rcl} \sigma^2 \text{ known} & \Longrightarrow & Z\text{-interval}, \\ \sigma^2 \text{ unknown} & \Longrightarrow & t\text{-interval}. \end{array}}$$

For finite samples, replacing σ by S changes the exact distribution from normal to Student's t.

4.5 Confidence Interval for the Difference of Two Normal Means

Suppose we have two independent samples

$$X_1, \ldots, X_{n_1} \overset{\text{iid}}{\sim} N(\mu_1, \sigma_1^2),$$

and

$$Y_1, \ldots, Y_{n_2} \overset{\text{iid}}{\sim} N(\mu_2, \sigma_2^2).$$

The parameter of interest is

$$\mu_1 - \mu_2.$$

Since

$$\mathbb{E}[\overline{X} - \overline{Y}] = \mu_1 - \mu_2,$$

the natural point estimator is

$$\overline{X} - \overline{Y}.$$

4.5.1 Variances Known

If σ_1^2 and σ_2^2 are known, then

$$\overline{X} - \overline{Y} \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right).$$

Hence

$$Z = \frac{(\overline{X} - \overline{Y}) - (\mu_1 - \mu_2)}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}} \sim N(0, 1).$$

Therefore the $(1 - \alpha)$ confidence interval for

$$\mu_1 - \mu_2$$

is

$$\boxed{\left[\overline{X} - \overline{Y} - z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}},\ \overline{X} - \overline{Y} + z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}\right].}$$

4.5.2 Variances Unknown but Equal

Suppose

$$\sigma_1^2 = \sigma_2^2 = \sigma^2$$

but σ^2 is unknown.

The two sample variances can then be combined into a pooled estimator.

Definition 4.7 (Pooled Sample Variance)**.** The pooled sample variance is

$$\boxed{S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}.}$$

The corresponding statistic is

$$T = \frac{(\overline{X}-\overline{Y}) - (\mu_1-\mu_2)}{S_p\sqrt{\dfrac{1}{n_1}+\dfrac{1}{n_2}}},$$

and

$$T \sim t_{n_1+n_2-2}.$$

Therefore the $(1-\alpha)$ confidence interval is

$$\boxed{\left[\overline{X}-\overline{Y} - t_{\alpha/2,n_1+n_2-2}S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}},\ \overline{X}-\overline{Y} + t_{\alpha/2,n_1+n_2-2}S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}\right].}$$

Remark 4.8. The pooled variance formula is appropriate only under the assumption

$$\sigma_1^2 = \sigma_2^2.$$

If the population variances are unknown and not assumed equal, the pooled t interval should not be used.

4.6 Confidence Interval for a Normal Variance

Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2),$$

and the parameter of interest is

$$\sigma^2.$$

4.6.1 Mean Known

If μ is known, then

$$\sum_{i=1}^{n} \frac{(X_i-\mu)^2}{\sigma^2} \sim \chi_n^2.$$

Hence

$$\mathbb{P}\left(\chi_{1-\alpha/2,n}^2 \le \frac{\sum_{i=1}^n (X_i-\mu)^2}{\sigma^2} \le \chi_{\alpha/2,n}^2\right) = 1-\alpha.$$

Solving for σ^2 gives

$$\boxed{\left[\frac{\sum_{i=1}^n (X_i-\mu)^2}{\chi_{\alpha/2,n}^2},\ \frac{\sum_{i=1}^n (X_i-\mu)^2}{\chi_{1-\alpha/2,n}^2}\right].}$$

4.6.2 Mean Unknown

If μ is unknown, then

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

Therefore,

$$\mathbb{P}\left(\chi^2_{1-\alpha/2,n-1} \leq \frac{(n-1)S^2}{\sigma^2} \leq \chi^2_{\alpha/2,n-1}\right) = 1-\alpha.$$

Solving for σ^2 gives

$$\boxed{\left[\frac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}}\right].}$$

Remark 4.9. The degrees of freedom change because estimating the unknown mean consumes one degree of freedom:

$$\boxed{\begin{array}{rcl} \mu \text{ known} & \Longrightarrow & n \text{ degrees of freedom,} \\ \mu \text{ unknown} & \Longrightarrow & n-1 \text{ degrees of freedom.} \end{array}}$$

Remark 4.10. Unlike the confidence interval for a mean, the confidence interval for σ^2 is generally not symmetric around S^2 because the chi-square distribution is not symmetric.

4.7 Confidence Interval for the Ratio of Two Normal Variances

Suppose

$$X_1, \ldots, X_{n_1} \overset{\text{iid}}{\sim} N(\mu_1, \sigma_1^2),$$

and

$$Y_1, \ldots, Y_{n_2} \overset{\text{iid}}{\sim} N(\mu_2, \sigma_2^2),$$

with the two samples independent.

We want to estimate

$$\frac{\sigma_1^2}{\sigma_2^2}.$$

Since

$$\frac{(n_1-1)S_1^2}{\sigma_1^2} \sim \chi^2_{n_1-1},$$

and

$$\frac{(n_2-1)S_2^2}{\sigma_2^2} \sim \chi^2_{n_2-1},$$

their standardized ratio satisfies

$$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{n_1-1,n_2-1}.$$

Equivalently,

$$\frac{S_1^2}{S_2^2}\frac{\sigma_2^2}{\sigma_1^2} \sim F_{n_1-1,n_2-1}.$$

Therefore,

$$\mathbb{P}\left(F_{1-\alpha/2;n_1-1,n_2-1} \le \frac{S_1^2}{S_2^2}\frac{\sigma_2^2}{\sigma_1^2} \le F_{\alpha/2;n_1-1,n_2-1}\right) = 1-\alpha.$$

Solving for

$$\frac{\sigma_1^2}{\sigma_2^2},$$

we obtain

$$\boxed{\left[\frac{S_1^2/S_2^2}{F_{\alpha/2;n_1-1,n_2-1}},\ \frac{S_1^2/S_2^2}{F_{1-\alpha/2;n_1-1,n_2-1}}\right].}$$

Using the reciprocal identity

$$F_{1-\alpha;\nu_1,\nu_2} = \frac{1}{F_{\alpha;\nu_2,\nu_1}},$$

the lower endpoint may also be written as

$$\frac{S_1^2}{S_2^2}F_{\alpha/2;n_2-1,n_1-1}.$$

Remark 4.11. The F distribution is asymmetric, so the resulting confidence interval for the ratio of variances is also asymmetric.

4.8 Large-Sample Confidence Intervals

Exact confidence intervals depend on exact sampling distributions such as the normal, t, chi-square, or F distributions.

When the exact distribution of an estimator is unavailable, a large-sample approximation can often be used.

4.8.1 General Asymptotic Normal Interval

Suppose an estimator $\widehat{\theta}$ satisfies approximately

$$\widehat{\theta} \approx N\left(\theta, \operatorname{Var}(\widehat{\theta})\right)$$

for large n.

Then approximately,

$$\frac{\widehat{\theta}-\theta}{\operatorname{SE}(\widehat{\theta})} \approx N(0,1).$$

Thus an approximate $(1-\alpha)$ confidence interval is

$$\boxed{\widehat{\theta} \pm z_{\alpha/2}\operatorname{SE}(\widehat{\theta}).}$$

Remark 4.12. This is the general large-sample pattern:

$$\boxed{\text{estimate} \pm \text{normal critical value} \times \text{estimated standard error.}}$$

Unlike the exact intervals developed earlier, its coverage probability is approximately rather than exactly equal to $1-\alpha$.

4.8.2 Large-Sample Interval Based on an MLE

Under suitable regularity conditions, the maximum likelihood estimator is asymptotically normal:

$$\sqrt{n}\left(\widehat{\theta}_{\text{MLE}} - \theta\right) \xrightarrow{\text{d}} N\left(0, \frac{1}{I_1(\theta)}\right).$$

Equivalently,

$$\widehat{\theta}_{\text{MLE}} \approx N\left(\theta, \frac{1}{I_n(\theta)}\right),$$

where

$$I_n(\theta) = nI_1(\theta).$$

Since θ is unknown, we usually replace it by

$$\widehat{\theta}_{\text{MLE}}$$

inside the Fisher information.

Thus an approximate confidence interval is

$$\boxed{\widehat{\theta}_{\text{MLE}} \pm z_{\alpha/2} \frac{1}{\sqrt{I_n(\widehat{\theta}_{\text{MLE}})}}.}$$

4.9 Large-Sample Confidence Interval for a Population Proportion

Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p).$$

Let

$$X = \sum_{i=1}^{n} X_i \sim \text{Binomial}(n, p),$$

and define the sample proportion

$$\widehat{p} = \frac{X}{n} = \overline{X}.$$

Then

$$\mathbb{E}[\widehat{p}] = p,$$

and

$$\text{Var}(\widehat{p}) = \frac{p(1-p)}{n}.$$

By the central limit theorem,

$$\frac{\widehat{p} - p}{\sqrt{p(1-p)/n}} \approx N(0, 1).$$

Replacing the unknown p in the standard error by $\widehat{p}$ gives

$$\text{SE}(\widehat{p}) \approx \sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}.$$

Therefore a commonly used approximate confidence interval is

$$\boxed{\widehat{p} \pm z_{\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}.}$$

Remark 4.13. This interval is based on the normal approximation to the binomial distribution and is therefore a large-sample interval.

Its accuracy may be poor when n is small or when p is very close to 0 or 1.

4.10 Sample Size Determination

Confidence interval formulas can also be used in reverse to determine how large a sample is needed to achieve a desired precision.

4.10.1 Normal Mean with Known Variance

For a normal mean with known variance, the margin of error is

$$E = z_{\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

If we require

$$E \le d,$$

then

$$z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le d.$$

Therefore,

$$\boxed{n \ge \left(\frac{z_{\alpha/2}\sigma}{d}\right)^2.}$$

Since n must be an integer, we round upward.

4.10.2 Population Proportion

For a population proportion, the approximate margin of error is

$$E = z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}.$$

Hence

$$n \ge \frac{z_{\alpha/2}^2 p(1-p)}{d^2}.$$

If no prior estimate of p is available, use

$$p(1-p) \le \frac{1}{4}.$$

Thus the conservative choice is

$$\boxed{n \ge \frac{z_{\alpha/2}^2}{4d^2}.}$$

Remark 4.14. The value

$$p = \frac{1}{2}$$

maximizes

$$p(1-p),$$

so it gives the largest required sample size and therefore provides a conservative design.

4.11 Summary of Confidence Intervals

For a normal sample,

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2),$$

the principal confidence intervals are:

$$\boxed{\mu,\ \sigma^2 \text{ known}: \qquad \overline{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}}$$

$$\boxed{\mu,\ \sigma^2 \text{ unknown}: \qquad \overline{X} \pm t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}}$$

$$\boxed{\sigma^2,\ \mu \text{ unknown}: \qquad \left[\frac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}},\ \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}}\right]}$$

and for two independent normal populations,

$$\boxed{\mu_1 - \mu_2,\ \sigma_1^2, \sigma_2^2 \text{ known}:}$$

$$\boxed{(\overline{X} - \overline{Y}) \pm z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

while, if the unknown variances are assumed equal,

$$\boxed{(\overline{X} - \overline{Y}) \pm t_{\alpha/2,n_1+n_2-2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}.$$

For the ratio of two normal variances,

$$\boxed{\left[\frac{S_1^2/S_2^2}{F_{\alpha/2;n_1-1,n_2-1}},\ \frac{S_1^2/S_2^2}{F_{1-\alpha/2;n_1-1,n_2-1}}\right].}$$

The general principle behind all of these formulas is

$$\boxed{\text{sampling distribution} \longrightarrow \text{pivotal quantity} \longrightarrow \text{probability statement} \longrightarrow \text{confidence interval.}}$$

Chapter 5

Hypothesis Testing

5 Hypothesis Testing

5.1 Introduction to Hypothesis Testing

Statistical hypothesis testing provides a formal procedure for deciding whether observed data are compatible with a specified claim about an unknown population parameter.

Suppose

$$X = (X_1, \ldots, X_n)$$

is a random sample from a distribution depending on a parameter

$$\theta \in \Theta.$$

A hypothesis testing problem divides the parameter space into two parts.

Definition 5.1 (Null and Alternative Hypotheses)**.** A statistical hypothesis testing problem is written as

$$H_0 : \theta \in \Theta_0 \qquad \text{versus} \qquad H_1 : \theta \in \Theta_1,$$

where

$$\Theta_0 \cap \Theta_1 = \varnothing.$$

The hypothesis H_0 is called the *null hypothesis*, and H_1 is called the *alternative hypothesis*.

For example,

$$H_0 : \mu = \mu_0 \qquad \text{versus} \qquad H_1 : \mu \neq \mu_0$$

is a two-sided test, whereas

$$H_0 : \mu = \mu_0 \qquad \text{versus} \qquad H_1 : \mu > \mu_0$$

is a one-sided test.

5.2 Critical Region and Decision Rule

Let

$$\mathcal{X}$$

denote the sample space of

$$X = (X_1, \ldots, X_n).$$

A hypothesis test divides $\mathcal{X}$ into two regions.

Definition 5.2 (Critical Region)**.** A subset

$$C \subseteq \mathcal{X}$$

is called the *critical region* or *rejection region* if

$$X \in C \quad \Longrightarrow \quad \text{reject } H_0.$$

Its complement

$$C^c$$

is called the acceptance region or non-rejection region.

Thus the decision rule is

$$\boxed{\begin{array}{lll} X \in C & \Longrightarrow & \text{reject } H_0, \\ X \notin C & \Longrightarrow & \text{do not reject } H_0. \end{array}}$$

Remark 5.3. In modern statistical language, it is usually better to say

"do not reject H_0"

rather than

"accept H_0".

Failure to reject H_0 does not prove that H_0 is true; it only means that the observed data do not provide sufficiently strong evidence against it.

5.3 Type I and Type II Errors

Because the decision is based on random data, errors are unavoidable.

Definition 5.4 (Type I Error)**.** A *Type I error* occurs when

$$H_0$$

is true but is rejected.

Its probability is

$$\boxed{\alpha = \mathbb{P}(\text{reject } H_0 \mid H_0 \text{ true}).}$$

Definition 5.5 (Type II Error)**.** A *Type II error* occurs when

$$H_0$$

is false but is not rejected.

Its probability is commonly denoted by

$$\boxed{\beta = \mathbb{P}(\text{do not reject } H_0 \mid H_1 \text{ true}).}$$

The two possible errors can be summarized as

	Do not reject H_0	Reject H_0
H_0 true	Correct	Type I error
H_1 true	Type II error	Correct

Remark 5.6. For a fixed sample size, decreasing the probability of Type I error usually makes it harder to reject H_0 and therefore tends to increase the probability of Type II error.

Thus α and β generally cannot both be made arbitrarily small without increasing the sample size.

5.4 Power Function, Size, and Significance Level

Definition 5.7 (Power Function)**.** For a rejection region C, the power function of a test is

$$\boxed{\pi(\theta) = \mathbb{P}_\theta(X \in C).}$$

Thus

$$\pi(\theta) = \mathbb{P}_\theta(\text{reject } H_0).$$

If

$$\theta \in \Theta_1,$$

then

$$\pi(\theta) = 1 - \beta(\theta).$$

Hence a larger power means that the test is more likely to detect a false null hypothesis.

Definition 5.8 (Size of a Test)**.** The size of a test is

$$\boxed{\sup_{\theta \in \Theta_0} \pi(\theta).}$$

A test is said to have significance level α if

$$\pi(\theta) \leq \alpha \qquad \text{for all } \theta \in \Theta_0.$$

For a simple null hypothesis

$$H_0 : \theta = \theta_0,$$

a size-α test satisfies

$$\mathbb{P}_{\theta_0}(X \in C) = \alpha.$$

Remark 5.9. The significance level controls the probability of falsely rejecting the null hypothesis.

The power controls the probability of correctly rejecting the null hypothesis when the alternative is true.

A good test should therefore have

$$\boxed{\text{small Type I error} \qquad \text{and} \qquad \text{large power.}}$$

5.5 The p-Value

The rejection-region approach fixes a significance level α first. An equivalent and very common approach reports the strength of evidence through a p-value.

Definition 5.10 (p-Value)**.** The p-value is the probability, calculated under H_0, of observing a test statistic at least as extreme as the observed value in the direction specified by H_1.

The decision rule is

$$\boxed{p\text{-value} \leq \alpha \quad \Longrightarrow \quad \text{reject } H_0.}$$

Otherwise,

$$p\text{-value} > \alpha \quad \Longrightarrow \quad \text{do not reject } H_0.$$

Remark 5.11. A p-value is *not*

$$\mathbb{P}(H_0 \text{ is true} \mid \text{data}).$$

In frequentist hypothesis testing, the null hypothesis is treated as fixed. The p-value is calculated from the distribution of the data under H_0.

5.6 Neyman–Pearson Lemma

We first consider testing two simple hypotheses:

$$H_0 : \theta = \theta_0 \qquad \text{versus} \qquad H_1 : \theta = \theta_1.$$

Among all tests with the same significance level, we would like to choose the one having the largest power under H_1.

Definition 5.12 (Most Powerful Test)**.** A size-α test is called *most powerful* for testing

$$H_0 : \theta = \theta_0$$

against

$$H_1 : \theta = \theta_1$$

if it has the largest power at θ_1 among all tests of size at most α.

Theorem 5.13 (Neyman–Pearson Lemma)**.** *Consider the simple hypotheses*

$$H_0 : \theta = \theta_0 \qquad \textit{versus} \qquad H_1 : \theta = \theta_1.$$

A most powerful test of size α rejects H_0 for sufficiently large values of the likelihood ratio

$$\boxed{\frac{L(\theta_1; x)}{L(\theta_0; x)}.}$$

Equivalently, the critical region has the form

$$\boxed{C = \left\{ x : \frac{L(\theta_1; x)}{L(\theta_0; x)} \geq k \right\},}$$

where k is chosen so that

$$\mathbb{P}_{\theta_0}(X \in C) = \alpha.$$

Remark 5.14. The interpretation is natural.

If

$$\frac{L(\theta_1; x)}{L(\theta_0; x)}$$

is large, then the observed sample is much more likely under H_1 than under H_0.

Therefore large likelihood ratios provide evidence against H_0.

5.7 Example of the Neyman–Pearson Lemma

Example 5.15 (Exponential Distribution)**.** Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Exponential}(\lambda),$$

with density

$$f(x; \lambda) = \lambda e^{-\lambda x}, \qquad x > 0.$$

Consider

$$H_0 : \lambda = \lambda_0 \qquad \text{versus} \qquad H_1 : \lambda = \lambda_1,$$

where

$$\lambda_1 > \lambda_0.$$

The likelihood ratio is

$$\begin{aligned}\frac{L(\lambda_1; x)}{L(\lambda_0; x)} &= \frac{\lambda_1^n e^{-\lambda_1 \sum x_i}}{\lambda_0^n e^{-\lambda_0 \sum x_i}} \\ &= \left(\frac{\lambda_1}{\lambda_0}\right)^n \exp\left\{-(\lambda_1 - \lambda_0)\sum_{i=1}^n x_i\right\}.\end{aligned}$$

Since

$$\lambda_1 - \lambda_0 > 0,$$

the likelihood ratio decreases as

$$\sum_{i=1}^n X_i$$

increases.

Therefore the most powerful test rejects H_0 for sufficiently small values of

$$\sum_{i=1}^n X_i.$$

Hence the rejection region has the form

$$C = \left\{\sum_{i=1}^n X_i \le c\right\},$$

where c is determined by the significance level.

5.8 Likelihood Ratio Test

The Neyman–Pearson lemma applies directly only when both H_0 and H_1 are simple hypotheses.

The likelihood ratio test extends the likelihood principle to general hypotheses.

Suppose

$$H_0 : \theta \in \Theta_0 \qquad \text{versus} \qquad H_1 : \theta \in \Theta \setminus \Theta_0.$$

Definition 5.16 (Likelihood Ratio Statistic)**.** The likelihood ratio statistic is

$$\boxed{\Lambda(X) = \frac{\sup_{\theta \in \Theta_0} L(\theta; X)}{\sup_{\theta \in \Theta} L(\theta; X)}.}$$

Since

$$\Theta_0 \subseteq \Theta,$$

we always have

$$0 \le \Lambda(X) \le 1.$$

The numerator is the largest likelihood attainable while H_0 is imposed, whereas the denominator is the largest likelihood without that restriction.

Thus small values of

$$\Lambda(X)$$

indicate that imposing H_0 substantially reduces the likelihood.

The likelihood ratio test therefore rejects for

$$\boxed{\Lambda(X) \le c.}$$

Remark 5.17. There are two reciprocal conventions for likelihood ratios.

The Neyman–Pearson form often uses

$$\frac{L(\theta_1)}{L(\theta_0)}$$

and rejects for *large* values.

The general likelihood ratio statistic is usually defined as

$$\Lambda = \frac{\sup_{\Theta_0} L}{\sup_{\Theta} L}$$

and rejects for *small* values.

These conventions are equivalent after taking reciprocals.

5.9 Procedure for Constructing an LRT

A likelihood ratio test can usually be constructed using the following steps.

1. Compute the unrestricted maximum likelihood estimator
$$\widehat{\theta} = \arg\max_{\theta \in \Theta} L(\theta).$$
2. Compute the restricted maximum likelihood estimator
$$\widehat{\theta}_0 = \arg\max_{\theta \in \Theta_0} L(\theta).$$
3. Form
$$\Lambda(X) = \frac{L(\widehat{\theta}_0; X)}{L(\widehat{\theta}; X)}.$$
4. Simplify $\Lambda(X)$ as a function of an appropriate test statistic
$$T(X).$$
5. Determine the rejection region from the distribution of $T(X)$ under H_0.
6. Choose the critical value so that the test has the required significance level α.

5.10 Asymptotic Distribution of the Likelihood Ratio

For complicated models, the exact distribution of Λ may not be easy to obtain.

Under suitable regularity conditions,

$$\boxed{-2 \log \Lambda \xrightarrow{\text{d}} \chi^2_r,}$$

where r is the number of independent restrictions imposed by H_0.

This result provides a general large-sample likelihood ratio test.

Remark 5.18. For example, if the full model has two free parameters and H_0 fixes one independent parameter relation, then typically

$$r = 1.$$

This asymptotic chi-square result is separate from Pearson's chi-square goodness-of-fit statistic introduced later, although both lead to chi-square limiting distributions.

5.11 Tests for a Normal Mean

Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2).$$

We consider testing hypotheses about μ.

5.12 One-Sample Z Test: Variance Known

Suppose σ^2 is known and consider

$$H_0 : \mu = \mu_0.$$

Under H_0,

$$Z = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1).$$

This is the test statistic.

5.12.1 Upper-Tailed Test

For

$$H_0 : \mu = \mu_0 \qquad \text{versus} \qquad H_1 : \mu > \mu_0,$$

reject H_0 if

$$\boxed{Z \geq z_\alpha.}$$

Equivalently,

$$\boxed{\overline{X} \geq \mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}}.}$$

The p-value is

$$\boxed{p = 1 - \Phi(z_{\text{obs}}).}$$

5.12.2 Lower-Tailed Test

For

$$H_0 : \mu = \mu_0 \qquad \text{versus} \qquad H_1 : \mu < \mu_0,$$

reject if

$$\boxed{Z \leq -z_\alpha.}$$

The p-value is

$$\boxed{p = \Phi(z_{\text{obs}}).}$$

5.12.3 Two-Sided Test

For

$$H_0 : \mu = \mu_0 \qquad \text{versus} \qquad H_1 : \mu \neq \mu_0,$$

reject if

$$\boxed{|Z| \geq z_{\alpha/2}.}$$

Equivalently,

$$Z \leq -z_{\alpha/2} \qquad \text{or} \qquad Z \geq z_{\alpha/2}.$$

The p-value is

$$\boxed{p = 2\left[1 - \Phi(|z_{\text{obs}}|)\right].}$$

5.13 One-Sample t Test: Variance Unknown

Now suppose σ^2 is unknown.

Under

$$H_0 : \mu = \mu_0,$$

the statistic

$$\boxed{T = \frac{\overline{X} - \mu_0}{S/\sqrt{n}}}$$

satisfies

$$T \sim t_{n-1}.$$

For the upper-tailed alternative

$$H_1 : \mu > \mu_0,$$

reject if

$$\boxed{T \geq t_{\alpha,n-1}.}$$

For the lower-tailed alternative

$$H_1 : \mu < \mu_0,$$

reject if

$$\boxed{T \leq -t_{\alpha,n-1}.}$$

For the two-sided alternative

$$H_1 : \mu \neq \mu_0,$$

reject if

$$\boxed{|T| \geq t_{\alpha/2,n-1}.}$$

Remark 5.19. The exact one-sample t test requires the population to be normal.

The reason for using the t distribution rather than the standard normal distribution is that σ has been replaced by the random quantity S.

5.14 Two-Sample Test for Normal Means

Suppose two independent random samples are observed:

$$X_1, \ldots, X_{n_1} \overset{\text{iid}}{\sim} N(\mu_1, \sigma_1^2)$$

and

$$Y_1, \ldots, Y_{n_2} \overset{\text{iid}}{\sim} N(\mu_2, \sigma_2^2).$$

We want to test

$$H_0 : \mu_1 - \mu_2 = \delta_0.$$

5.14.1 Variances Known

If σ_1^2 and σ_2^2 are known, then under H_0,

$$\boxed{Z = \frac{(\overline{X} - \overline{Y}) - \delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0,1).}$$

The rejection regions are obtained exactly as in the one-sample Z test.

5.14.2 Variances Unknown but Equal

Suppose

$$\sigma_1^2 = \sigma_2^2 = \sigma^2,$$

where σ^2 is unknown.

Define the pooled sample variance

$$\boxed{S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}.}$$

Then

$$\boxed{T = \frac{(\overline{X} - \overline{Y}) - \delta_0}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1+n_2-2}}$$

under H_0.

For

$$H_1 : \mu_1 - \mu_2 \neq \delta_0,$$

reject H_0 if

$$\boxed{|T| \geq t_{\alpha/2, n_1+n_2-2}.}$$

Remark 5.20. The pooled t test requires the assumption

$$\boxed{\sigma_1^2 = \sigma_2^2.}$$

The pooled variance combines the two independent estimates of the same population variance.

5.15 Connection Between Confidence Intervals and Tests

There is a close relationship between two-sided hypothesis tests and confidence intervals.

Consider

$$H_0 : \theta = \theta_0 \qquad \text{versus} \qquad H_1 : \theta \neq \theta_0.$$

For many standard procedures,

$$\boxed{\text{Reject } H_0 \quad \Longleftrightarrow \quad \theta_0 \notin (1-\alpha) \text{ confidence interval.}}$$

Example 5.21. For a normal mean with known variance, the $(1-\alpha)$ confidence interval is

$$\overline{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

The hypothesis

$$H_0 : \mu = \mu_0$$

is rejected at level α exactly when

$$\mu_0$$

lies outside this interval.

5.16 Chi-Square Goodness-of-Fit Test

The previous tests concerned unknown numerical parameters such as a mean.

We now consider a different question:

$$\boxed{\text{Does the entire population distribution agree with a specified model?}}$$

This leads to the chi-square goodness-of-fit test.

Suppose the sample space is divided into m mutually exclusive categories

$$A_1, \ldots, A_m,$$

with

$$A_i \cap A_j = \varnothing, \qquad i \neq j.$$

Let

$$p_j = \mathbb{P}(X \in A_j), \qquad j = 1, \ldots, m,$$

so that

$$\sum_{j=1}^{m} p_j = 1.$$

Given a random sample

$$X_1, \ldots, X_n,$$

let N_j denote the number of observations falling in category A_j.

Then

$$\boxed{(N_1, \ldots, N_m) \sim \text{Multinomial}(n; p_1, \ldots, p_m).}$$

Moreover,

$$\sum_{j=1}^{m} N_j = n.$$

5.17 Pearson's Chi-Square Statistic

Under a completely specified null distribution, the expected number of observations in category A_j is

$$E_j = np_j.$$

The difference

$$N_j - np_j$$

measures the discrepancy between the observed and expected frequencies.

Pearson's statistic is

$$\boxed{Q = \sum_{j=1}^{m} \frac{(N_j - np_j)^2}{np_j}.}$$

Remark 5.22. The numerator

$$(N_j - np_j)^2$$

measures squared discrepancy, while division by

$$np_j$$

standardizes the discrepancy according to its expected scale.

Thus Q becomes large when the observed frequencies differ substantially from those predicted by H_0.

5.18 Completely Specified Distribution

Suppose

$$H_0 : p_j = p_{j0}, \qquad j = 1, \dots, m,$$

where all probabilities

$$p_{10}, \dots, p_{m0}$$

are completely known.

Under H_0 and appropriate large-sample conditions,

$$\boxed{Q = \sum_{j=1}^{m} \frac{(N_j - np_{j0})^2}{np_{j0}} \xrightarrow{\text{d}} \chi^2_{m-1}.}$$

Therefore the level-α test rejects H_0 if

$$\boxed{Q \geq \chi^2_{\alpha, m-1}.}$$

Remark 5.23. There are m observed counts

$$N_1, \dots, N_m,$$

but they satisfy the constraint

$$N_1 + \cdots + N_m = n.$$

Thus only $m - 1$ counts can vary freely, which explains the $m - 1$ degrees of freedom.

5.19 Example: Discrete Uniform Distribution

Example 5.24 (Testing a Discrete Uniform Distribution)**.** Suppose a random variable can take the values

$$1, 2, 3, 4, 5,$$

and we want to test

$$H_0 : \mathbb{P}(X = j) = \frac{1}{5}, \qquad j = 1, \ldots, 5.$$

If the sample size is n, then the expected frequency in every category is

$$E_j = \frac{n}{5}.$$

For observed frequencies

$$N_1, \ldots, N_5,$$

the test statistic is

$$Q = \sum_{j=1}^{5} \frac{(N_j - n/5)^2}{n/5}.$$

Under H_0,

$$Q \approx \chi^2_4.$$

At significance level α, reject the discrete uniform model if

$$Q \geq \chi^2_{\alpha,4}.$$

5.20 Goodness-of-Fit with Estimated Parameters

The null distribution is not always completely specified.

For example, we may test whether the population follows a Poisson distribution

$$H_0 : X \sim \text{Poisson}(\lambda),$$

while λ itself is unknown.

In this case, λ must first be estimated from the sample.

Suppose s independent parameters are estimated from the data. Then, under standard regularity conditions,

$$\boxed{Q \approx \chi^2_{m-s-1}.}$$

Remark 5.25. The degrees of freedom are

$$\boxed{m - 1 - s.}$$

The term $m - 1$ comes from the constraint

$$N_1 + \cdots + N_m = n,$$

while each estimated parameter imposes one additional constraint and therefore removes one further degree of freedom.

5.21 Example: Testing a Poisson Distribution

Example 5.26 (Poisson Goodness-of-Fit Test)**.** Suppose

$$X_1, \dots, X_n$$

are observations of count data and we wish to test

$$H_0 : X \sim \text{Poisson}(\lambda)$$

for some unknown $\lambda > 0$.

The MLE of λ is

$$\widehat{\lambda} = \overline{X}.$$

Under the fitted model,

$$\widehat{p}_j = \mathbb{P}_{\widehat{\lambda}}(X \in A_j).$$

The expected frequency in category A_j is

$$E_j = n\widehat{p}_j.$$

Therefore Pearson's statistic becomes

$$\boxed{Q = \sum_{j=1}^{m} \frac{(N_j - E_j)^2}{E_j}.}$$

Since one parameter λ has been estimated,

$$s = 1,$$

and hence

$$\boxed{Q \approx \chi^2_{m-2}.}$$

Large values of Q provide evidence that the Poisson model does not fit the observed data.

5.22 Combining Categories

The chi-square approximation requires the expected category frequencies not to be too small.

If some expected frequencies

$$E_j = np_j$$

are very small, adjacent categories should be combined whenever this is meaningful.

A commonly used practical guideline is to avoid categories with very small expected counts, often using approximately

$$E_j \geq 5$$

as a rule of thumb.

Remark 5.27. If categories are combined, the number m of categories changes and the degrees of freedom must be recalculated using the final number of categories.

5.23 Pearson Statistic as a Measure of Discrepancy

Pearson's statistic can be written as

$$Q = \sum_{j=1}^{m} \frac{(\text{Observed}_j - \text{Expected}_j)^2}{\text{Expected}_j}.$$

Thus the goodness-of-fit procedure has the intuitive form

Compare observed frequencies with model-predicted frequencies.

If these differences are small,

$$Q$$

is small and the data are compatible with the proposed model.

If the differences are large,

$$Q$$

is large and the null model is rejected.

5.24 Summary of Hypothesis Testing

The general structure of hypothesis testing is

Specify H_0 and H_1

$\Downarrow$

Choose a test statistic $T(X)$

$\Downarrow$

Find its distribution under H_0

$\Downarrow$

Choose significance level α

$\Downarrow$

Determine the rejection region

$\Downarrow$

Calculate the observed statistic or p-value

$\Downarrow$

Reject or do not reject H_0.

For normal means, the main test statistics are

$$\boxed{Z = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)}$$

when σ is known, and

$$\boxed{T = \frac{\overline{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}}$$

when σ is unknown.

For two independent normal populations with unknown but equal variances,

$$\boxed{T = \frac{(\overline{X} - \overline{Y}) - \delta_0}{S_p\sqrt{\dfrac{1}{n_1} + \dfrac{1}{n_2}}} \sim t_{n_1+n_2-2}.}$$

For a multinomial goodness-of-fit problem,

$$\boxed{Q = \sum_{j=1}^{m} \frac{(N_j - E_j)^2}{E_j}}$$

has approximately a chi-square distribution with

$$\boxed{m - 1}$$

degrees of freedom when all probabilities are known, and

$$\boxed{m - s - 1}$$

degrees of freedom when s parameters are estimated from the sample.

Finally, the central principles of optimal testing are

$$\boxed{\text{Neyman–Pearson lemma} \Longrightarrow \text{most powerful test for simple hypotheses,}}$$

and

$$\boxed{\text{Likelihood ratio} \Longrightarrow \text{general method for constructing hypothesis tests.}}$$

Acknowledgements

These notes were developed from my study of *Mathematical Statistics*, with the main structure and core material based on the textbook written by Professor Guoliang Tian, who taught my course in mathematical statistics.

The original notes began as a concise collection of important concepts, formulas, and frequently used results extracted from Professor Tian's textbook during my study of the course. I have subsequently reorganized these materials into a more systematic set of notes, expanding brief statements with additional explanations, derivations, examples, and connections between related statistical ideas.

I would like to express my sincere gratitude to Professor Guoliang Tian for his teaching and for the textbook on which much of the mathematical content and organization of these notes is based. His presentation of probability, sampling distributions, estimation, confidence intervals, and hypothesis testing provided the foundation for my study of mathematical statistics.

Any additional explanations, reorganizations, or errors in these notes are my own.